# Toward a four-dimensional Yang–Mills construction

Notebook 32: determinant composition and finite coupling tuning

Research working document

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**Abstract**

We identify the actual leading quantum correction and the precise drift input needed for finite ultraviolet tuning. All statements use the composite ordinary constraint, its pivot coordinates, the absolute contour penalty and the retained Gaussian of the preceding construction.

First, the finite classical Maxwell leakage tends uniformly to zero along a cofinal nested period family, even when the preceding homogeneous depth varies. The actual finite quantum map is jointly continuous in its interaction class and its positive real coupling; the continuity argument requires no continuous choice of small representatives.

Second, we derive an explicit convergent trace formula for the Maxwell coefficient of the pivot and Gaussian determinants. Global color invariance removes the second stationary response only after the full scalar chain rule is taken. The constraint, current, gauge and logarithmic-chart curvature terms remain in the precision derivatives. At a fixed regulator, exact Haar pushforward composition and finite-dimensional Laplace asymptotics prove a cocycle for these actual leading determinants. Exact infinite classical Maxwell preservation then makes their coefficients additive across scales.

An exact finite gauge and coarea identity also rewrites the leading correction as vector, based-ghost and coarse-constraint determinants. The coarse factors remain explicit. A flat Rayleigh test shows that the based ghost alone has no uniform gap at the final physical scale, so a reference comparison must retain the coupled cancellations. The coarse constraint covariance is identified with a shifted actual effective-Hessian inverse. For a fixed sufficiently large positive shift, its Maxwell contribution is uniformly bounded in depth and volume.

Finally, under an explicitly unproved positive averaged drift hypothesis for the actual finite leading coefficients, a continuous shooting argument tunes arbitrarily long finite local histories to one fixed positive weak coupling. The auxiliary clipping is proved absent on the selected histories. This is a conditional tuning theorem, not an evaluation of the drift. A uniform positive block margin in every varying-depth structural limit would suffice: qualitative compactness transfers that margin to the actual finite regulators without a convergence rate.

The remaining one-loop task is a bound on the difference between the actual composite coefficient and an explicitly normalized reference logarithm, uniform in depth. Mixed and large-field histories, continuum observables, nontriviality and reconstruction remain part of the full existence argument.

The objective remains a nontrivial four-dimensional Yang–Mills quantum theory. This notebook continues the actual homogeneous local construction of Notebooks 15–31. Its finite local density identities do not by themselves represent the entire Wilson measure. Results and conditional inputs are distinguished in the statements below.

## 1 Uniform disappearance of finite classical marginal leakage

This section gives a qualitative uniform statement along the standard nested compatible period family. It does not assert exact finite marginal preservation or an exponential convergence rate. The preceding classical depth may vary arbitrarily with the period. The proof uses the already established exact

preservation of the true infinite Maxwell coefficient in each joint classical system, rather than identifying different limiting systems.

### 1.1 Period family and statement

Fix the primitive ratio $\ell$, an admissible composite ratio $R = \ell^s$, auxiliary grouping $J$, carrier exponent $a_0$, and one positive common analytic radius $b$. All these data are fixed. Let $\mathcal{P}$ be a compatible nested family of cubic output periods tending to infinity and satisfying

$$\text{for each fixed } q \geq 0, \quad R^q \mid N \quad \text{for all sufficiently late } N \in \mathcal{P}. \tag{1}$$

All grouping and source-geometry divisibilities are also imposed. The standard nested source tori have this property; for example $N = 2\ell^k$ does when the fixed grouping is compatible. Arbitrary growing multiples of $J$ with bounded $\ell$-adic valuation are not covered by this argument.

Let $\mathcal{T}_{M,R,N}$ be the actual pre-gauge classical substitution from the input torus of current period $RN$ to the output torus of period $N$. Use the actual finite short-lift Maxwell ratios, each normalized by its own actual finite homogeneous reference, and define

$$\Lambda_{M,R,N} = \beta_{MR,N}\mathcal{T}_{M,R,N} - \beta_{M,RN}. \tag{2}$$

This is a linear functional on the invariant finite activity quotient at the common $a_0, b$. On an extracted input it is the finite classical marginal leakage.

**Theorem 1.1** (Qualitative uniform finite leakage)**.** *For all sufficiently large lower period thresholds $K$, set*

$$\omega_{R,J,b}(K) = \sup_{\substack{M,\ N\in\mathcal{P}\\ N\geq K}} \sup_{\substack{x \text{ invariant}\\ \|x\|\leq b^2}} |\Lambda_{M,R,N}x|. \tag{3}$$

*Then*

$$\omega_{R,J,b}(K) \longrightarrow 0 \quad (K \longrightarrow \infty). \tag{4}$$

*The supremum allows every permitted preceding homogeneous depth $M$, with no fixed-depth or bounded-depth restriction.*

The function in (3) is finite and decreasing above the reference thresholds. Indeed the uniform finite moment bound and fixed-$R$ transport bound give

$$\omega_{R,J,b}(K) \leq C_\beta(B_R + 1). \tag{5}$$

The factor $b^{-2}$ in the dual moment norm cancels the input normalization $b^2$. The theorem asserts a qualitative improvement of this bound, not a rate.

### 1.2 Compactness of varying finite input classes

Take any sequence $M_n$, periods $N_n \in \mathcal{P}$ tending to infinity, and invariant finite input classes $x_n$ with $\|x_n\| \leq b^2$. Choose representatives $E_n$ on the input tori with

$$\|E_n\|_{a_0,b} \leq b^2 + \epsilon_n, \qquad \epsilon_n \downarrow 0. \tag{6}$$

Average each family over the finite translations by $J$-cells, translating its carriers and arguments together. Its anchored norm does not increase, because the translations permute the root sums. Its normalized total remains exactly the same by the total translation invariance of $x_n$. The resulting array is covariant under $J$-cell translations, while the total retains all original unit-translation, point and local gauge identities. No coefficientwise gauge invariance is needed.

Apply the proved analytic-interaction compactness argument at the same open radius $b$. On every fixed embedded connected carrier, finite-dimensional Montel compactness gives local uniform convergence of its coefficient function. A countable diagonal extraction gives an infinite family $E_\infty$. Fatou on finite rooted carrier collections, followed by radii increasing to $b$, yields

$$\|E_\infty\|_{a_0,b} \le b^2. \tag{7}$$

There is no analytic-radius loss.

Vacuum-normalized totals converge on compactly supported fields. The anchored cardinality reserve makes the large-carrier remainder uniformly small, and every fixed carrier eventually embeds without wrapping. Local gauge and spatial identities pass on their common compact-field domains by the same total-continuity argument. Thus $E_\infty$ represents an invariant infinite class $x_\infty$.

Fixed-order total kernels converge locally, in particular

$$K_{2,n}^{\mathrm{in}} \longrightarrow K_2(x_\infty). \tag{8}$$

They have the common exponential row and moment bounds following from the activity norm. No numerical value of $M_n$ enters this input compactness argument.

### 1.3 Joint extraction when the classical depth varies

The classical maps and references must be extracted jointly with the input interactions. For every fixed depth shift $j \ge 0$ and endpoint power $L = R^q$, use the actual finite data at depth $M_n R^j$, on compatible periods obtained from $RN_n$ by finitely many divisions by $R$ and $L$. Condition (1) ensures that all these data exist for sufficiently large $n$, and their periods still tend to infinity.

For each fixed endpoint $L$, an observed current output fiber contains only $L^4$ times a fixed number of link and color coordinates. This dimension is independent of the preceding refinement $M_n$. The previously proved classical inputs are uniform in that refinement:

- a positive common holomorphic radius;
- strict support and cardinality bounds;
- complete $C^{1,\alpha}$ bounds and the exact $1/L$ pre-gauge amplitude;
- actual finite reference transport and minimum comparison identities;
- the controlled finite near-identity gauge comparisons.

Apply analytic compactness to the reference families and complete-output-family compactness to each fixed-endpoint map. One diagonal subsequence contains every depth shift and every endpoint power in this countable collection.

Denote the resulting references by $S_j^*$ and the maps by $\mathcal{T}_{j,q}^*$, with endpoint $R^q$. The index $j$ records a relative depth shift. No assertion $S_j^* = S_{M_* R^j}$ for a finite $M_*$ is made. The uniform flat Maxwell normalization and exponential kernel bounds give

$$\beta(S_j^*) = 1. \tag{9}$$

The fixed-radius reference bounds, exact transport and scalar semigroup pass to this joint system. For compact fields the anchored tails justify passage of normalized reference values, without passage of an extensive vacuum energy. The scalar semigroup on every invariant input class uses the actual finite small connecting gauges and their endpoint estimates. These estimates are uniform in $M_n$; their

compact extraction and the summable-gauge argument therefore apply to this varying-depth sequence as well. They never require the numerical label $M_n$ to be fixed.

Equivalently, for each fixed endpoint pair, pass the finite comparison germ together with its actual controlled gauge data, and then use holomorphic continuation on the common field domain. The comparison germ can be smaller for a large endpoint; the uniform pre-gauge domain remains the same. The support estimate at fixed classical reserve is uniform in all endpoint powers, and no high-jet field-dilation radius is used here.

The resulting system satisfies

$$\mathcal{T}^*_{j,q}S^*_j = S^*_{j+q}, \qquad \mathcal{T}^*_{j+q,r}\mathcal{T}^*_{j,q} = \mathcal{T}^*_{j,q+r}. \tag{10}$$

It also has the uniform quadratic-only spatial estimate at fixed $b$ for arbitrarily large endpoints $R^q$. It therefore satisfies precisely the hypotheses of the exact infinite Maxwell-preservation proof, which gives

$$\beta(\mathcal{T}^*_{0,1}x) = \beta(x) \quad \text{for every invariant } x. \tag{11}$$

This is a varying-depth structural limit, not an invocation of a fixed-$M$ limit theorem. Different limiting systems need not coincide. Every possible system extracted this way has (11), which is the only property needed below.

### 1.4 The transported Hessian and its moment tails

Let

$$H_n = Da_{M_n,R,N_n}(0) \tag{12}$$

be the actual linearized pre-gauge angle map. It takes output current-angle directions on period $N_n$ to input current-angle directions on period $RN_n$. Its transpose in the scalar Hessian formula is the ordinary counting-coordinate transpose: all physical rescaling is already in this angle map.

The complete support-family estimate and Cauchy at zero give

$$|H_n(b,c)| \le C_R e^{-\kappa_R d(\mathrm{root}(b),c)}. \tag{13}$$

The root is the final-unit cell containing the input bond after division of its coordinates by $R$. Fixed internal link and color conventions are absorbed into $C_R$. One may retain the amplitude $1/R$, but $R$ is fixed in this proof.

Each root contains only $O(R^4)$ input coordinates. Thus (13) gives both row and column exponential Schur bounds, with constants independent of $M_n$ and volume. Joint local holomorphic convergence implies local convergence $H_n \to H_*$.

The total first derivative of each input vanishes by global semisimple color invariance, and the actual pre-gauge map vanishes at zero. Consequently the exact finite Hessian chain rule is

$$K^{\mathrm{out}}_{2,n} = H_n^T K^{\mathrm{in}}_{2,n} H_n. \tag{14}$$

There is no term involving the second derivative of the background map.

The exponential envelopes for the two response legs and the input kernel give a uniform output exponential row bound. To see this, connect the two output source roots through the intermediate input roots. A fixed fraction of the minimum response and input decay exponent pays the output separation; the remaining geometric sums are bounded by their infinite-lattice sums. The $O(R^4)$ fiber multiplicity is fixed. No factor from $M_n^4$ fine coordinates appears.

These bounds also prove local convergence of (14): truncate its two intermediate root sums to a fixed neighborhood, pass the finite sum entrywise, and then remove the truncation using the common exponential majorant. Hence

$$K^{\mathrm{out}}_{2,*} = H_*^T K_2(x_\infty) H_*, \tag{15}$$

which is the Hessian of $\mathcal{T}^*_{0,1}x_\infty$. The same estimates give uniformly summable second displacement moments.

### 1.5 Convergence of the actual finite Maxwell ratios

For locally converging torus kernels with a uniform exponential row bound, the symmetric short-lift second moments converge to the true infinite second moment. First restrict to a fixed displacement radius; that finite sum converges locally. The remaining second moment is bounded uniformly by an exponentially decaying tail in that radius. The short-lift cutoff $\lfloor N_n/16 \rfloor$ tends to infinity, so it produces no residual term. Fixed midpoint shifts are handled by the same majorant.

The finite reference denominators tend uniformly to one by the established flat $q_M$ periodization estimate, including for varying $M_n$. Therefore

$$\begin{aligned} \beta_{M_n,RN_n}(x_n) &\longrightarrow \beta(x_\infty), \\ \beta_{M_nR,N_n}(\mathcal{T}_{M_n,R,N_n}x_n) &\longrightarrow \beta(\mathcal{T}_{0,1}^{*}x_\infty). \end{aligned} \tag{16}$$

These are the actual finite reference-normalized ratios. Their full nonlinear references have not been replaced by canonical periodizations, and no extensive total action limit is used.

### 1.6 Uniform contradiction and consequences

*Proof of Theorem 1.1.* If (4) failed, there would be $\delta > 0$ and sequences as above with

$$|\Lambda_{M_n,R,N_n}x_n| \geq \delta$$

for every $n$. Extract the joint subsequence constructed in the preceding sections. By (16), the limiting difference is

$$\beta(\mathcal{T}_{0,1}^{*}x_\infty) - \beta(x_\infty) = 0$$

by (11). This contradicts the lower bound and proves the theorem. □

On extracted inputs the conclusion reads

$$\sup_{\substack{M,\ N\in\mathcal{P},\ N\geq K \\ \beta_{M,RN}(x)=0,\ \|x\|\leq b^2}} |\beta_{MR,N}(\mathcal{T}_{M,R,N}x)| \longrightarrow 0. \tag{17}$$

The unextracted statement (3) is slightly stronger and needs no projection-loss factor. By linear homogeneity it gives, for arbitrary invariant inputs,

$$|\Lambda_{M,R,N}x| \leq \omega_{R,J,b}(K)b^{-2}\|x\|, \qquad N \in \mathcal{P},\ N \geq K. \tag{18}$$

For a prescribed per-step drift tolerance, choose $K$ so that $\omega_{R,J,b}(K)$ is sufficiently small. Every step of a permitted finite history whose output period remains at least $K$ then has the same leakage bound, uniformly in the preceding depth and the length of that history. On the established envelope $\|x\| \leq C_{\mathrm{int}}b^2$, multiply the tolerance by $C_{\mathrm{int}}$, or choose $K$ so that $\omega_{R,J,b}(K)$ is the desired tolerance divided by this fixed constant.

### 1.7 Qualifications

The result is qualitative. It supplies no exponential rate, no summability of leakage over infinitely many changing periods, and no exact zero at finite volume. The per-step consequence is a magnitude bound, not a sign or an accumulated-drift assertion.

The cofinal divisibility assumption (1) is part of the theorem. An extension to arbitrary increasing compatible multiples with bounded endpoint divisibility would require an additional uniform period-extension or local-volume comparison. It does not follow merely from $N \to \infty$.

The limiting systems have not been identified with one another. Exact infinite Maxwell preservation holds in each separately and rules out every nonzero leakage subsequence. No quantum drift sign, determinant value, infinite physical running trajectory, or global measure interpretation follows from this compactness argument.

## 2 Continuity in the real coupling at a finite regulator

This section concerns the actual homogeneous local macrostep and its specified common-domain continuation. It supplies continuity needed for a finite endpoint argument. It proves no coupling sign, endpoint crossing, or identification with the full Wilson pushforward.

### 2.1 Fixed data and statement

Fix one allowed finite macrostep, including its preceding classical depth $M$, current and output tori, ratios $R, J$, common angle radius $b$, carrier exponent $a_0$, actual retained chart, and all strict common-domain budgets. Use the finite activity quotients by equality of vacuum-normalized totals on the full product ball. Their norms are infima of anchored family norms.

Let $I$ be a compact interval contained in $(0, g_0]$, where $g_0$ meets the common-domain and two-input comparison bounds. For example, take the continuous threshold

$$T(g) = \max\{T_0, A\sqrt{\log(1/g)}\}. \tag{19}$$

More generally, any positive continuous threshold satisfying those bounds suffices. Only continuity and boundedness on $I$ will be used. Denote the actual finite quotient map by

$$F_g(x) = T_{\mathrm{cl}}x + \Delta + \mathcal{Q}_{\mathrm{det}} + \mathcal{L}_g(Q_x). \tag{20}$$

Here $T_{\mathrm{cl}}$ is classical background transport, the next two terms are the fixed pivot and Gaussian determinant corrections, and $Q_x$ is the inherited fluctuation difference. The stationary backgrounds, precision and chart are independent of $g$. The coupling is inserted *inside* the nonlinear chart as $\Phi_V(g\xi)$.

**Theorem 2.1** (Finite real-coupling continuity)**.** *On the common admissible quotient ball, for instance $\|x\| \le \bar{\epsilon}/4$, the map $(g, x) \mapsto F_g(x)$ is jointly continuous for $0 < g \le g_0$. Both activity topologies use the same radius $b$ and exponent $a_0$. Consequently the actual extracted update*

$$(\gamma, x) \longmapsto \Big(\gamma - \beta_{\mathrm{out}}(F_{\gamma^{-1/2}}(x)), P_{\mathrm{out}}F_{\gamma^{-1/2}}(x)\Big) \tag{21}$$

*is continuous on its admissible positive weak-coupling domain. The finite maps $\beta_{\mathrm{out}}, P_{\mathrm{out}}$ are normalized by the actual finite reference and do not depend on $g$.*

The proof permits constants depending exponentially on the finite volume. It asserts neither a volume-uniform modulus nor continuity at $g = 0$. In particular it does not require a continuous choice of near-minimal activity representatives.

### 2.2 Strict inherited-field margin

Fix one input class $x$ and choose one admissible representative $E$, held fixed as $g$ varies; one may take $\|E\| < \bar{\epsilon}/2$. The proved actual joint-family construction gives

$$f_g(w, \xi) = \Omega_R(a_p(w), g\xi). \tag{22}$$

Its absolute rooted family bound supplies, under the strict budgets,

$$\|a_p(w)\|_\infty \le b/4, \qquad \|f_g(w, \xi)\|_\infty \le b/4 \tag{23}$$

for all $\|w\|_\infty < b$, $g \in I$, and every actual retained link/color window of threshold $T(g)$. Fixed norm conversions have already been included in the common-domain constants. The argument below only requires a fixed inner radius $sb$ with $s < 1$.

Write $E_{\mathrm{raw}}(A) = \sum_H E_H(A|_H)$, and let $V_{\mathrm{in}}$ be the number of input $J$-cells. The anchored norm implies

$$\sum_H \|E_H\|_b \le V_{\mathrm{in}} \|E\|_{a_0;b}. \tag{24}$$

Directional one-variable Cauchy on the product ball gives

$$|E_{\mathrm{raw}}(A) - E_{\mathrm{raw}}(B)| \le \frac{V_{\mathrm{in}} \|E\|_{a_0;b}}{(1-s)b} \|A - B\|_\infty, \qquad \|A\|_\infty, \|B\|_\infty \le sb. \tag{25}$$

Indeed the segment between $A, B$ stays in the inner product ball; each unit max-norm direction has a complex disk of radius at least $(1-s)b$. Apply Cauchy on any smaller disk, integrate the derivative along the segment, sum using (24), and pass to the limiting disk radius. Thus an arbitrary input need only be bounded and holomorphic on its original $b$ ball. No extension past that ball is assumed.

### 2.3 Fixing the real integration window

Let $D$ be the total finite number of real retained scalar coordinates. Let $\Xi_1 \subset \mathbb{R}^D$ be the product of the actual centered unit retained link/color balls. Its dilation $T(g)\Xi_1$ is exactly the actual window; its boundary is Lebesgue-null. Let $A(w)$ be the retained precision, with the fixed analytic square-root determinant branch. Changing real variables $\xi = T(g)\zeta$ gives the exact Gaussian expectation

$$\begin{aligned} Z_g(w) = \frac{\det A(w)^{1/2}}{(2\pi)^{D/2}} T(g)^D \int_{\Xi_1} \exp\{ &-\tfrac{1}{2} T(g)^2 \zeta^T A(w) \zeta + V_{\mathrm{base}}(w, T(g)\zeta; g) \\ &+ E_{\mathrm{raw}}(f_g(w, T(g)\zeta)) - E_{\mathrm{raw}}(a_p(w)) \} \, d\zeta. \end{aligned} \tag{26}$$

The background-only pivot and determinant terms may remain outside this expression as in (20). Coupling powers, compact gauge-insertion constants and flat Haar constants are separately tracked vacuum factors. No contour deformation or replacement by a whole-fiber Euclidean cutoff occurs. The chart argument is $\Phi_V(gT(g)\zeta)$.

The independent classical domains used in constructing the actual chart and base interaction give the following bounds on this fixed regulator. The precision and its determinant branch are bounded for all $\|w\|_\infty < b$. The local analytic coefficient maps are bounded and uniformly Lipschitz in their fluctuation arguments on a slightly larger fixed chart domain containing every $gT(g)\zeta$. Moreover $V_{\mathrm{base}}(w, T(g)\zeta; g)$ is uniformly bounded and uniformly continuous in $g \in I$, uniformly over the full open $b$ ball and $\zeta \in \Xi_1$.

The last assertion follows directly from the exact factors as well. The classical and gauge Taylor remainders are analytic functions of $g\xi$, divided by $g^2$ after removing the constant, linear and quadratic terms. The remaining Haar and Jacobian factors are analytic in the same argument. On a compact positive $g$ interval all these expressions have bounded parameter derivatives on the fixed strict chart margin. Their removable singularities at zero coupling are not needed. An anchored bound on a classical support series bounds its extensive sum at this fixed volume.

Since $a_p$ is independent of $g$, continuity of $gT(g)$ and these analytic bounds imply

$$\sup_{\substack{\|w\|_\infty < b \\ \zeta \in \Xi_1}} \left\| f_g(w, T(g)\zeta) - f_{g_*}(w, T(g_*)\zeta) \right\|_\infty \longrightarrow 0. \tag{27}$$

Use (25) for the inherited exponent. The Gaussian term has the same uniform continuity because $A(w)$ is bounded and $T$ is continuous. Every exponent in (26) has a common finite absolute bound; the prefactor is uniformly bounded and $\Xi_1$ has finite measure. Uniform domination therefore proves

$$\|Z_g - Z_{g_*}\|_{H^\infty(b)} \longrightarrow 0. \tag{28}$$

This is a supremum over the entire open $b$ ball, not merely convergence on compact subballs.

### 2.4 A lower bound on the full ball and logarithms

The strict Gaussian theorem supplies nonvanishing throughout the $b$ ball and a connected logarithm family with a common anchored bound $H_I$ for $g \in I$. If $V_{\mathrm{out}}$ is the number of output $J$-cells, its normalized total satisfies

$$\sup_{\|w\|_\infty < b} |\log Z_g(w) - \log Z_g(0)| \le 2V_{\mathrm{out}} H_I. \tag{29}$$

The bound is unaffected by raw logarithm constants. By (28) at $w = 0$ and nonvanishing, on a compact positive interval around $g_*$, $m_0 = \inf_{g\in I} |Z_g(0)| > 0$. Hence

$$\inf_{\substack{g\in I \\ \|w\|_\infty < b}} |Z_g(w)| \ge m_0 \exp(-2V_{\mathrm{out}} H_I) =: m_I > 0. \tag{30}$$

Nonvanishing alone would not yield such an infimum on the open ball; the uniform logarithm estimate is essential.

When $\|Z_g - Z_{g_*}\|_b < m_I/2$, use the convergent power-series logarithm

$$\ell_g(w) = \log\left(1 + \frac{Z_g(w) - Z_{g_*}(w)}{Z_{g_*}(w)}\right), \qquad \|\ell_g\|_b \le 2m_I^{-1}\|Z_g - Z_{g_*}\|_b. \tag{31}$$

If $l_g$ is any selected holomorphic logarithm of $Z_g$, then $l_g - l_{g_*} - \ell_g$ is constant on the connected product ball, because its exponential is one. Thus

$$\big\|[l_g - l_g(0)] - [l_{g_*} - l_{g_*}(0)]\big\|_b \le 4m_I^{-1}\|Z_g - Z_{g_*}\|_b \longrightarrow 0. \tag{32}$$

No continuity of a raw branch convention has been assumed.

### 2.5 From totals to the finite quotient norm

Let $H_{\mathrm{all}}$ be the full connected output $J$-cell torus. Assign any bounded holomorphic normalized total $h$ to this one carrier and zero to all others. This is an admissible finite activity representation and gives

$$\|[h]\|_{\mathrm{quot}} \le e^{a_0 V_{\mathrm{out}}}\|h\|_b. \tag{33}$$

If $h$ has invariant total, this representative retains that property; coefficientwise gauge invariance is unnecessary. The exterior terms in (20) are independent of $g$. Applying (33) to (32) proves, for fixed $x$,

$$\|F_g(x) - F_{g_*}(x)\|_{\mathrm{quot}} \longrightarrow 0. \tag{34}$$

The volume cost in (33) is intentional. It avoids an unsupported inference about the norm continuity of every canonical cluster coefficient or relocalized array.

### 2.6 Joint continuity and finite histories

The finite two-input estimate is uniform on $I$:

$$\|F_g(x) - F_g(y)\|_{\mathrm{quot}} \le \left[B_T + C_{\mathrm{quant}} \sup_{h\in I} \frac{hT(h)}{b}\right] \|x - y\|_{\mathrm{quot}}. \tag{35}$$

It follows from admissible near-minimal representatives $Y, D, X = Y + D$ and the stronger comparison budget $a_Q\bar{\epsilon} \le \eta_*/(128e)$. It does not require a continuous representative selection. For $(g, x) \to (g_*, x_*)$, split the difference as

$$F_g(x) - F_{g_*}(x_*) = F_g(x) - F_g(x_*) + F_g(x_*) - F_{g_*}(x_*).$$

Equations (35) and (34) make both terms tend to zero. Boundedness of the actual finite $\beta_{\mathrm{out}}$ and $P_{\mathrm{out}}$, together with continuity of $\gamma \mapsto \gamma^{-1/2}$, proves Theorem 2.1.

Every fixed finite composition of the extracted updates is continuous while all intermediate inputs and couplings remain in their common admissible domains. This follows by induction over the finite stages. One can work near a trajectory lying strictly within these domains, or use the finite-induction envelope to ensure admissibility on a prescribed parameter interval.

Near-minimal representatives implementing the history may jump with the initial coupling. Exact finite-total descent makes the quotient update and its beta increment independent of those choices. Raw vacuum factors remain separately tracked by their exact compensation identities; an endpoint statement about $\gamma$ does not require a continuous raw representative convention.

Continuity alone supplies no endpoint crossing. An intermediate-value argument also needs values on opposite sides of the desired endpoint and control keeping the intervening histories weak and admissible. A bound on the magnitude of each increment may keep a sufficiently high finite initial $\gamma$ weak, but it does not tune a fixed terminal coupling or establish an ultraviolet trajectory. The present conclusion remains finite-volume, positive-coupling, and confined to the specified local analytic map.

# 3 An explicit trace formula for the pivot and Gaussian drift

Fix the primitive ratio, a composite ratio $R$, the actual composite ordinary average, its selected pivots and the direct-$R$ internal contour gauge. The preceding homogeneous classical depth $M$ is arbitrary. All finite-regulator formulas in this section are exact. Their infinite-volume interpretation uses one chosen joint local-limit system with the previously proved uniform exponential bounds. No numerical value, sign, old-depth limit or universal beta coefficient is asserted.

## 3.1 The exact external factors

Use real Lie coordinates orthonormal for the normalized trace, and their bilinear complexification. Write current link angles as $a$ and final coarse link angles as $w$. Retained-coordinate matrix traces use the actual Euclidean counting coordinates of the Gaussian integral. They carry no extra old fine-volume factor.

Let $F_R$ be the actual composite ordinary group average. Its relative constraint and differential are

$$F_a(B) = \frac{1}{i}\log\left[F_R(e^{iB}e^{ia})F_R(e^{ia})^{-1}\right], \qquad C_a = D_B F_a(0). \tag{36}$$

For the unique selected pivot $p(c)$ of a final bond $c$,

$$(C_a)_{c,p(c)} = R^{1-d}\pi_c(a), \qquad \pi_c(0) = I. \tag{37}$$

The actual pivot insertion, retained insertion and chart satisfy

$$\begin{aligned} C_a h_a &= I, \qquad K_a = E_{\mathrm{free}} - h_a C_a E_{\mathrm{free}}, \\ D_z\Phi_a(0) &= K_a, \qquad D_z^2\Phi_a(0)[u,v] = -h_a D_B^2 F_a(0)[K_a u, K_a v]. \end{aligned} \tag{38}$$

The chart solves $F_a(\Phi_a(z)) = 0$ and has retained coordinates exactly $z$.

Let $G_R$ be the actual absolute contour gauge function. It is not replaced by a background-relative penalty. Define

$$\mathcal{W}_M(a) = S_M(e^{ia}) + G_R(e^{ia}), \qquad f(a,z) = \frac{1}{i}\log[e^{i\Phi_a(z)}e^{ia}], \tag{39}$$

so $f(a,0) = a$, and set

$$\mathscr{A}_M(a) = D_z^2[\mathcal{W}_M(f(a,z))]_{z=0}. \tag{40}$$

This is the full analytic precision on the independent background ball. Its Gaussian need not be centered at an arbitrary independent background.

The actual physical branch $a_*(w)$ satisfies

$$a_*(0) = 0, \qquad F_R(e^{ia_*(w)}) = e^{iw}, \qquad P_R(e^{ia_*(w)}; y, x) = I.$$

It is the centered branch of the integral. Put

$$A(w) = \mathscr{A}_M(a_*(w)), \qquad A_0 = \mathscr{A}_M(0), \qquad C_0 = A_0^{-1}. \tag{41}$$

The exact external logarithms are

$$\Delta(w) = -\sum_c \log\det \pi_c(a_*(w)), \qquad \mathcal{Q}_{\det}(w) = -\frac{1}{2}\log\det[A(w)A_0^{-1}], \qquad D_M = \Delta + \mathcal{Q}_{\det}. \tag{42}$$

All logarithmic branches in this display vanish at zero. The factor $R^{1-d}$ cancels between each pivot determinant and its zero-background value.

The signs follow from the inverse pivot Jacobian and the Gaussian factor $\det A^{-1/2}$. There is no additional background Haar factor $j(a_*)$: the exact link Haar coordinates are left-relative and give $j(\Phi_a(g\xi))$, whose logarithm vanishes at $g = 0$. The fluctuation straightening ratio is also one at $\xi = 0$. Powers of $g$, the gauge-insertion constant and flat Gaussian constants are field independent and remain outside $D_M$. These definitions retain the actual mean, composite map, chart and gauge; a direct-$R$ replacement of the constraint or an axial covariance would give a different formula.

### 3.2 The complete finite-regulator chain rule

For final coarse directions $u, v$, write

$$H_u = Da_*(0)u, \qquad B_{uv} = D^2a_*(0)[u, v]. \tag{43}$$

Define independent-background vertices by

$$\begin{aligned} \mathscr{A}_1[p] &= D\mathscr{A}_M(0)[p], & \mathscr{A}_2[p, q] &= D^2\mathscr{A}_M(0)[p, q], \\ \pi_{c,1}[p] &= D\pi_c(0)[p], & \pi_{c,2}[p, q] &= D^2\pi_c(0)[p, q]. \end{aligned} \tag{44}$$

The physical derivatives are

$$\begin{aligned} A_u &= \mathscr{A}_1[H_u], & A_{uv} &= \mathscr{A}_2[H_u, H_v] + \mathscr{A}_1[B_{uv}], \\ \pi_{c,u} &= \pi_{c,1}[H_u], & \pi_{c,uv} &= \pi_{c,2}[H_u, H_v] + \pi_{c,1}[B_{uv}]. \end{aligned} \tag{45}$$

**Proposition 3.1** (Exact finite trace Hessian)**.** *For the actual finite integral,*

$$\begin{aligned} D^2D_M(0)[u, v] = &-\sum_c \operatorname{tr}\{\pi_{c,uv} - \pi_{c,u}\pi_{c,v}\} \\ &-\frac{1}{2}\operatorname{Tr}\{C_0A_{uv} - C_0A_uC_0A_v\}. \end{aligned} \tag{46}$$

*Here lowercase trace acts on one pivot color block and uppercase trace acts on all retained coordinates.*

*Proof.* Differentiate the logarithmic determinants in (42), using

$$D\log\det A[B] = \operatorname{Tr}(A^{-1}B), \qquad D(A^{-1})[B] = -A^{-1}BA^{-1},$$

and $\pi_c(0) = I$. Products need not commute. Cyclicity of the trace makes the answer symmetric in $u, v$. □

In particular, constrained stationarity does not say that the left current of $S_M$ vanishes. It does not permit deleting the constraint-curvature terms inside $\mathscr{A}_1$ or $\mathscr{A}_2$.

### 3.3 The second stationary response cancels after taking the scalar trace

Introduce the independent-background scalar

$$d_M(a) = -\sum_c \log\det \pi_c(a) - \frac{1}{2}\log\det[\mathscr{A}_M(a)A_0^{-1}]. \tag{47}$$

This function is not asserted to be invariant under arbitrary local background gauges. The constant-color symmetry suffices for the next lemma.

**Lemma 3.2** (Semisimple cancellation of the first derivative)**.** *For a semisimple Lie algebra,*

$$Dd_M(0) = 0. \tag{48}$$

*Consequently the full Hessian in (46) reduces to*

$$\begin{aligned} D^2D_M(0)[u,v] = & -\sum_c \operatorname{tr}\{\pi_{c,2}[H_u,H_v] - \pi_{c,1}[H_u]\pi_{c,1}[H_v]\} \\ & -\frac{1}{2}\operatorname{Tr}\{C_0\mathscr{A}_2[H_u,H_v] - C_0\mathscr{A}_1[H_u]C_0\mathscr{A}_1[H_v]\}. \end{aligned} \tag{49}$$

*Proof.* Let $O = \mathrm{Ad}_k$ in the fixed real orthonormal Lie basis. Endpoint covariance of the actual averages gives

$$\pi_c(Oa) = O\pi_c(a)O^{-1}. \tag{50}$$

The unique small pivot chart is equivariant because both its constraint and retained-coordinate prescription are. The absolute contour gauge is a sum of class functions, and the actual scalar action is invariant under global conjugation. Thus, with $O_{\mathrm{ret}}$ the direct sum on retained coordinates,

$$\mathscr{A}_M(Oa) = O_{\mathrm{ret}}\mathscr{A}_M(a)O_{\mathrm{ret}}^T, \qquad O_{\mathrm{ret}}^TO_{\mathrm{ret}} = I. \tag{51}$$

The flat precision commutes with this representation. Equations (50)–(51) prove invariance of (47), first on the real domain and then by holomorphic continuation with its zero-at-origin branches.

A semisimple Lie algebra has no invariant linear functional on one adjoint color slot. Apply the invariance to a variation on one current bond. Its linear functional is therefore zero. Doing this for every bond proves (48). The argument applies separately to the pivot sum and the Gaussian trace logarithm.

After substituting (45) into (46), the coefficient of $B_{uv}$ is precisely $Dd_M(0)[B_{uv}]$. This proves (49). □

This cancellation does not say that $\mathscr{A}_1 = 0$, $\pi_{c,1} = 0$, or that the current and chart-curvature operators vanish. For an abelian factor, retain (46) unless its linear functional has been separately shown to vanish.

### 3.4 The actual flat stationary lift

The first response appearing in (49) is explicit in the proved flat operators. Let

$$C_R = D_a\{\log F_R(e^{ia})/i\}_{a=0}, \qquad h = h_0, \quad K = K_0, \quad P_M = q_M + T_R^TT_R. \tag{52}$$

Here $q_M$ is the ordinary homogeneous classical Hessian, and $T_R$ is the linearized direct-$R$ internal contour map. The constraint in $C_R$ is still the composite map. The angle/counting normalization gives

$$A_0 = K^TP_MK, \qquad B_M = K^TP_Mh = K^Tq_Mh, \qquad H = h - KC_0B_M. \tag{53}$$

The simplification uses $T_R h = 0$, since internal contours use no pivots.

To verify the formula, use the identity-background affine chart

$$\log F_R(e^{ia})/i = C_R a + \mathcal{C}_R(a), \qquad \mathcal{C}_R(0) = D\mathcal{C}_R(0) = 0,$$
$$\mathfrak{a}(w,z) = Kz + hw - h\mathcal{C}_R(\mathfrak{a}(w,z)). \tag{54}$$

Put $\mathcal{T}_M(w,z) = \mathcal{W}_M(\mathfrak{a}(w,z))$. Since the full action is stationary at zero,

$$D_z\mathcal{T}_M(w,z) = A_0 z + B_M w + O((|w|+|z|)^2). \tag{55}$$

The positive flat retained Hessian and the established local uniqueness identify its analytic critical point with the physical branch. Thus $z_u = -C_0 B_M u$ and $H_u = hu + Kz_u$, proving (53). No $M^d$ or $R^d$ transpose factor belongs in these angle-coordinate matrices.

For completeness, the second response retained in the general formula is also explicit. Set $t_u = (u, z_u)$ and define $z_{uv}$ by the counting-pairing identity

$$\langle \zeta, A_0 z_{uv}\rangle = -D^3\mathcal{T}_M(0)[(0,\zeta), t_u, t_v] \quad \text{for every retained } \zeta. \tag{56}$$

Then

$$B_{uv} = -hD^2\mathcal{C}_R(0)[H_u, H_v] + Kz_{uv}. \tag{57}$$

These equations follow by twice differentiating the retained stationarity equation and (54); they are available even when the color cancellation is not.

### 3.5 The complete precision vertices

Set

$$E_a = D_z f(a,0), \qquad T_a = D_z^2 f(a,0). \tag{58}$$

With

$$J_a X = \int_0^1 \mathrm{Ad}(e^{ita})X\, dt, \qquad \Xi_a(B) = \frac{1}{i}\log[e^{iB}e^{ia}],$$

the exact tensors are

$$E_a = J_a^{-1}K_a,$$
$$T_a[u,v] = -J_a^{-1}h_a D_B^2 F_a(0)[K_a u, K_a v] + D_B^2\Xi_a(0)[K_a u, K_a v]. \tag{59}$$

Consequently the arbitrary-background precision is

$$\mathscr{A}_M(a)[u,v] = D^2\mathcal{W}_M(a)[E_a u, E_a v] + D\mathcal{W}_M(a)[T_a(u,v)]. \tag{60}$$

The two terms of $T_a$ are respectively nonlinear constraint curvature and conversion from the left-relative exponential coordinate to the ordinary link logarithm. The absolute gauge function remains inside $\mathcal{W}_M$, so its nonphysical background curvature is included as well. Its gradient vanishes on the physical branch, but this does not permit replacing the independent analytic family by a relative quadratic gauge form.

Write $W_n = D^n\mathcal{W}_M(0)$, so that $W_1 = 0$, $W_2 = P_M$, $E_0 = K$. Define

$$E_1[p] = D_a E_0[p], \qquad E_2[p,q] = D_a^2 E_0[p,q], \qquad T_0 = T_{a=0}, \qquad T_1[p] = D_a T_0[p].$$

For retained directions $r, s$, direct differentiation gives

$$\mathscr{A}_1[p][r,s] = W_3[p, Kr, Ks] + W_2[E_1[p]r, Ks] + W_2[Kr, E_1[p]s]$$
$$+ W_2[p, T_0(r,s)]. \tag{61}$$

The last term is a first-background-order current and constraint contribution; it remains despite $W_1 = 0$. The second vertex is

$$\begin{aligned}\mathscr{A}_2[p,q][r,s] = {} & W_4[p,q,Kr,Ks] \\ & + W_3[p,E_1[q]r,Ks] + W_3[p,Kr,E_1[q]s] \\ & + W_3[q,E_1[p]r,Ks] + W_3[q,Kr,E_1[p]s] \\ & + W_2[E_2[p,q]r,Ks] + W_2[Kr,E_2[p,q]s] \\ & + W_2[E_1[p]r,E_1[q]s] + W_2[E_1[q]r,E_1[p]s] \\ & + W_3[p,q,T_0(r,s)] \\ & + W_2[p,T_1[q](r,s)] + W_2[q,T_1[p](r,s)]. \end{aligned} \tag{62}$$

Only the $W_1 D_a^2 T$ term disappears at zero. The pivot vertices follow by twice differentiating

$$\pi_c(a) = R^{d-1}(D_B F_a(0))_{c,p(c)}. \tag{63}$$

All chart operations here use the actual finite-dimensional logarithm and mean, and have finite range at the fixed $R$.

### 3.6 The classical cubic and quartic vertices

The classical vertices in (61)–(62) can be evaluated from the previously proved simultaneous corrected-slice equations. This avoids an unproved bound for the ordinary rooted average's physical adjoint.

Use the fine extension $\mathcal{H}_M$, constrained Green operator $\mathcal{C}_M$, flat coarse Hessian $q_M$, corrected constraint remainder $\mathcal{R}_M$, Wilson jet $f_\eta$, and its finite jet map $\mathcal{S}$. With the physical fine transposes and measures, the exact pair $Y = (A, m)$ obeys

$$\begin{aligned} A &= \mathcal{H}_M a - \mathcal{H}_M \mathcal{R}_M(A) - \mathcal{C}_M \mathcal{S}^T f_\eta(\mathcal{S}A) + \mathcal{C}_M D\mathcal{R}_M(A)^T m, \\ m &= q_M a - q_M \mathcal{R}_M(A) + (\mathcal{S}\mathcal{H}_M)^T f_\eta(\mathcal{S}A) - \mathcal{H}_M^T D\mathcal{R}_M(A)^T m. \end{aligned} \tag{64}$$

Write $Y = \mathcal{L}_M a + \mathcal{V}_M(Y)$, where $\mathcal{L}_M = (\mathcal{H}_M, q_M)$ and $\mathcal{V}_M(0) = D\mathcal{V}_M(0) = 0$. For $\mathcal{V}_n = D^n \mathcal{V}_M(0)$, its responses are

$$\begin{aligned} Y_1[u] &= \mathcal{L}_M u, \\ Y_2[u,v] &= \mathcal{V}_2[\mathcal{L}_M u, \mathcal{L}_M v], \\ Y_3[u,v,t] &= \mathcal{V}_3[\mathcal{L}_M u, \mathcal{L}_M v, \mathcal{L}_M t] \\ &\quad + \mathcal{V}_2[Y_2[u,v], \mathcal{L}_M t] + \mathcal{V}_2[Y_2[u,t], \mathcal{L}_M v] + \mathcal{V}_2[Y_2[v,t], \mathcal{L}_M u]. \end{aligned} \tag{65}$$

These are derivatives, not Taylor coefficients, so no factorial is omitted. The scalar envelope $DS_M(a) = m(a)$ and equality of the corrected and ordinary action values yield

$$\begin{aligned} D^3 S_M(0)[u,v,t] &= \langle (Y_2[u,v])_m, t \rangle, \\ D^4 S_M(0)[u,v,t,r] &= \langle (Y_3[u,v,t])_m, r \rangle. \end{aligned} \tag{66}$$

Add the explicit local derivatives of $G_R$ to obtain $W_3, W_4$.

These are finite compositions of the proved propagators and finite-range analytic vertices. The uniform estimates use complete physical cell spaces and smooth supported reconstruction, not unweighted sums over $M^4$ microscopic coordinates. The fine $\eta^4$ measure is already in the physical transposes of (64). It must not be inserted again into the retained-coordinate traces.

### 3.7 Absolute trace convergence and local kernels

The strict precision theorem supplies symmetric pieces with matrix rows and columns in their carrier and an analytic bound of the following form, at fixed positive $\sigma, a, \rho$:

$$\sup_i \sum_{Y \ni i} \sum_{j \in Y} e^{\sigma\tau(Y)+a|Y|} \|(\mathscr{A}_M - A_0)_{Y,ij}\|_\rho \le N. \tag{67}$$

This is uniform in preceding depth and permitted volume. The precision-composition theorem supplies the same form after stationary substitution, keeping the actual matrix rows rooted in their original carrier. The flat covariance has an exponential Schur bound. The pivot scalar and stationary field have their established strict and rooted bounds.

**Lemma 3.3** (Source-rooted trace bounds)**.** *On a fixed smaller analytic domain, the actual Hessian kernel*

$$\mathcal{K}_{xy} = D^2 D_M(0)[e_x, e_y]$$

*has an exponentially summable row bound*

$$\sup_x \sum_y e^{md(x,y)} |\mathcal{K}_{xy}| \le C_D \tag{68}$$

*for some $m > 0$. The traces in both (46) and (49) are absolutely convergent on compact or summable variation directions. Constants may depend on the fixed geometry, color and radii, but not on preceding depth or volume.*

*Proof.* A derivative in one background coordinate $x$ acts only on carriers containing $x$. Cauchy bounds it by a fixed constant times $\rho^{-1}$ times the original majorant. Two derivatives require both source slots in the carrier and give $\rho^{-2}$.

Reduce the spatial exponent in (67). For each matrix row $i$, the unused exponent pays $e^{-\delta d(i,x)}$, since $\tau(Y) \ge d(i,x)$ when $i, x \in Y$. Its sum over $i$ is the finite lattice bound $S_\delta$, also uniformly on periodic graphs. With a second source $y$, spend another exponent portion on $\tau(Y) \ge d(x,y)$.

For the composed precision, this yields, after fixed exponent reductions,

$$\sum_{ij} e^{m_1[d(i,x)+d(j,x)]} \|(A_x)_{ij}\| \le C_1,$$
$$\sum_{ij} \|(A_{xy})_{ij}\| \le C_2 e^{-m_1 d(x,y)}. \tag{69}$$

For the first line, use $d(i,x)+d(j,x) \le 2\tau(Y)$ and reserve an additional exponent for the row summation. Finite fiber and color factors are absorbed into the constants.

The same argument before composition, followed by the exponentially decaying external lift $H$, bounds the separate terms of (49). Thus the proof does not use cancellation of non-absolutely convergent quantities.

The trace with $A_{xy}$ is bounded by its absolute-entry sum and the covariance entry envelope. For the other trace expand

$$\mathrm{Tr}(C_0 A_x C_0 A_y) = \sum_{i,j,k,l} \mathrm{tr}\{C_{0,ij}(A_x)_{jk} C_{0,kl}(A_y)_{li}\}. \tag{70}$$

The source-root bounds and covariance envelope make this absolutely convergent. The $x$-root weight at $j$, the covariance weight between $j$ and $i$, and the $y$-root weight at $i$ jointly pay $d(x,y)$. All remaining factors are absolutely summable. Hence

$$|\,\mathrm{Tr}(C_0 A_x C_0 A_y)| \le C_3 e^{-m_2 d(x,y)}$$

for a smaller positive $m_2$. The pivot traces have the same bound by local dependence before composition and the rooted stationary family, or directly by their composed strict scalar expansion. A final exponent reduction allows the row sum in (68). □

Grouping retained coordinates into auxiliary $J$-cells only regroups these traces; it inserts no extra factor $J^d$ into the exact identities.

On the infinite lattice a total log determinant at a noncompact background need not exist as an extensive number. The displayed formulas concern local kernels and derivatives on compact or summable directions. They are defined by the absolutely convergent sums above or by a chosen joint local limit of the actual finite matrices and vertices. The common majorants permit that limit. No arbitrary finite activity is identified with the periodization of an independently selected infinite one.

### 3.8 A convergent Maxwell moment of the traces

Assume the chosen infinite total has the established translation, hypercubic and local gauge identities, and fix one simple color factor. The midpoint Hessian symbol then has

$$\widehat{\mathcal{K}}^{ab}_{\mu\nu}(p) = \delta^{ab}\beta(D_M)(|p|^2\delta_{\mu\nu} - p_\mu p_\nu) + O(|p|^4). \tag{71}$$

For $\mu \neq \nu$, choose one trace-orthonormal color vector $t^a$, and let $e_{x,\mu,a}$ be its final unit-spacing link direction. The midpoint offsets cancel for the diagonal orientation $\mu,\mu$. Therefore

$$\boxed{\beta(D_M) = -\frac{1}{2}\sum_{r\in\mathbb{Z}^4} r_\nu^2 D^2 D_M(0)[e_{0,\mu,a}, e_{r,\mu,a}].} \tag{72}$$

There is no color sum here. Hypercubic and color symmetry make the choices equivalent. The minus sign comes from twice differentiating $e^{ip\cdot r}$; the factor $1/2$ extracts the coefficient of $p_\nu^2$. The actual classical reference has coefficient one in this normalization.

Writing $H_r = He_{r,\mu,a}$, substitution of (49) gives

$$\begin{aligned}\beta(D_M) = \frac{1}{2}\sum_r r_\nu^2 \sum_c \operatorname{tr}\{\pi_{c,2}[H_0,H_r] - \pi_{c,1}[H_0]\pi_{c,1}[H_r]\} \\ + \frac{1}{4}\sum_r r_\nu^2 \operatorname{Tr}\{C_0\mathscr{A}_2[H_0,H_r] - C_0\mathscr{A}_1[H_0]C_0\mathscr{A}_1[H_r]\}.\end{aligned} \tag{73}$$

Every sum is absolutely convergent by Lemma 3.3. Truncating the external displacement at distance $L$ has error at most $C_{m'}e^{-m'L}$ for any $0 < m' < m$. Internal trace tails likewise have exponential bounds after anchoring the sources and spending a spare exponent.

At an actual finite regulator, the trace Hessian formulas remain exact. Apply the chosen symmetric short-lift moment and divide by the moment of the actual finite reference to obtain its finite Maxwell ratio. That reference denominator is $1 + O(e^{-cN})$, not automatically exactly one. No finite Ward identity at discrete momenta is interpolated to obtain (71); its use requires the stated infinite kernel and symmetries, or the separately justified local-limit passage.

### 3.9 What has and has not been reduced

The leading zero-input drift is expressed by actual local pivot vertices, the flat stationary lift (53), the retained covariance, and the full precision vertices (61)–(62). The classical cubic and quartic vertices have the finite operator recursion (64)–(66). All signs and coordinate normalizations are fixed.

The second physical stationary response was included before being removed by the proved semisimple color identity. The physical current was not replaced by an independent parameter, and neither chart-curvature contribution was omitted.

Evaluating (73), comparing its values as $M$ grows, determining a sign or a universal coefficient, and controlling a physical coupling trajectory remain further tasks. Absolute convergence of the trace expression alone proves none of those conclusions.

# 4 An exact gauge and coarea bridge for the actual determinant

This section rewrites the actual leading correction as coupled vector, ghost and coarse-constraint determinants. It keeps the constraint-current curvature and every finite orbit-density factor. It also exhibits a soft mode that prevents a separate uniformly massive estimate for the based ghost. The bounded finite-part comparison with a standard background-Feynman determinant remains a separate estimate.

## 4.1 The actual multiplier Hessian

Fix a finite compatible regulator and endpoint $K = \ell^k$. Let $U = V_*^{M,K}(W)$ be the physical stationary representative, so $F_K(U) = W$ and $P_K(U) = I$. All matrices use the actual current-link angle and scalar-site counting coordinates in a real trace-orthonormal color basis.

In left-relative coordinates define

$$s(x) = S_M(e^{ix}U), \qquad f(x) = \frac{1}{i}\log\Big[F_K(e^{ix}U)W^{-1}\Big], \qquad C = f'(0). \tag{74}$$

The target coordinate is a relative increment at $W$, not its ordinary logarithm. Constrained stationarity gives a unique multiplier $\lambda$ with $s'(0) = C^T\lambda$. Put

$$\mathcal{L} = s''(0) - \sum_c \lambda_c f_c''(0). \tag{75}$$

This is the Hessian of the critical Lagrangian $s - \lambda \cdot f$. Its second term is the full constraint-current curvature.

Let $C_p$ be the invertible actual pivot block, $I_p$ the pivot insertion, and $E$ the retained insertion. Write

$$h = I_p C_p^{-1}, \qquad K_0 = E - hCE, \qquad Ch = I, \qquad CK_0 = 0. \tag{76}$$

The subscript distinguishes $K_0$ from the endpoint ratio $K$. The nonlinear source chart satisfies

$$\Phi'(0) = K_0, \qquad \Phi''(0)[u,v] = -hf''(0)[K_0u, K_0v]. \tag{77}$$

Consequently

$$D_z^2 s(\Phi(z))\big|_{z=0} = K_0^T \mathcal{L} K_0.$$

The actual absolute contour penalty vanishes at the saddle; its Hessian is $T^TT$, where $T$ differentiates the contour logarithms in the established normalization. Thus

$$A_T = K_0^T(\mathcal{L} + T^TT)K_0 > 0. \tag{78}$$

Apart from field-independent constants, the actual leading amplitude is

$$|\det C_p|^{-1}\det(A_T)^{-1/2}. \tag{79}$$

The real determinant signs are fixed from the flat convention. All logarithms below are normalized at $W = 1$ and then continued on their chosen holomorphic branches.

### 4.2 The based gauge nullspace

Let $D_b$ send scalar site parameters vanishing at every $K$-center to their left-relative infinitesimal gauge variations at $U$. Exact based invariance of the ordinary constraint gives

$$CD_b = 0. \tag{80}$$

The scalar $s - \lambda \cdot f$ is based-gauge invariant and is critical at zero. Differentiating its gauge identity there gives

$$\mathcal{L}D_b = 0. \tag{81}$$

This need not hold for $s''$ without its multiplier-curvature term. The actual contour endpoint covariance gives

$$TD_b = -I \tag{82}$$

with the standard orientation. Changing the generator sign changes only this sign. A fixed color normalization in the gauge penalty contributes only a background-independent factor, canceled by normalization.

The based action is free, hence $D_b$ is injective and

$$G_b = D_b^T D_b > 0. \tag{83}$$

On $\ker C$, the form $\mathcal{L}$ is nonnegative, has nullspace precisely $\operatorname{ran} D_b$, and is positive on its quotient. Indeed, for any $x \in \ker C$, use (82) to add a unique based gauge vector making $Tx = 0$. Equation (81) leaves its $\mathcal{L}$ value unchanged. On this contour-zero complement, the positive form (78) equals $\mathcal{L}$. Equality occurs only for a vertical vector.

### 4.3 A finite Gaussian gauge-change identity

**Lemma 4.1** (Gauge replacement with its orbit density)**.** *Let $B_0$ be a symmetric nonnegative form on a finite space $Z$, with nullspace* $\operatorname{ran} V$*, positive on the quotient. Suppose $F_1, F_2 : Z \to \mathbb{R}^s$ have invertible $F_j V$. Then*

$$|\det(F_1 V)| \det(B_0 + F_1^T F_1)^{-1/2} = |\det(F_2 V)| \det(B_0 + F_2^T F_2)^{-1/2}. \tag{84}$$

*Proof.* Choose one complement $E_0$ to $\operatorname{ran} V$ and set $z = E_0 u + V\alpha$. The change-of-variable Jacobian is the same in the two integrals. The $B_0$ term depends only on $u$. Integration over $\alpha$ yields $(2\pi)^{s/2}/|\det(F_j V)|$, independently of $u$, by translation and an invertible linear change of variables. The remaining integral in $u$ is common. This proves (84); no orbit-coordinate density has been discarded. □

Apply the lemma on retained coordinates, with vertical insertion $V$ characterized by $K_0 V = D_b$. Take $F_1 = TK_0$ and $F_2 = D_b^T K_0$. Equations (82) and (83) give $F_1 V = -I$ and $F_2 V = G_b$. Therefore

$$\det(A_T)^{-1/2} = \det(G_b) \det(A_b)^{-1/2}, \qquad A_b = K_0^T(\mathcal{L} + D_b D_b^T)K_0. \tag{85}$$

This replaces the actual contour gauge by based covariant divergence. It does not yet replace it by a full-site massive background-Feynman gauge.

### 4.4 Coarea removes the pivot

Choose any symmetric coarse matrix $B$ such that

$$Q_b = \mathcal{L} + D_b D_b^T + C^T B C > 0, \qquad M_b = C Q_b^{-1} C^T > 0. \tag{86}$$

A completion exists at each fixed finite saddle: the first two terms are positive on $\ker C$, so a sufficiently large scalar $B$ makes the complementary Schur block positive. The condition is positivity of $Q_b$. Merely requiring $B > 0$ need not suffice; conversely $B$ itself may be indefinite.

Let $n$ and $m$ be the ambient and constraint dimensions. The same constrained Gaussian has the exact evaluations

$$\begin{aligned} \int_{\mathbb{R}^n} \delta(Cx) e^{-x^T Q_b x/2}\, dx &= (2\pi)^{(n-m)/2} |\det C_p|^{-1} \det(A_b)^{-1/2} \\ &= (2\pi)^{(n-m)/2} [\det Q_b \, \det M_b]^{-1/2}. \end{aligned} \tag{87}$$

For the first, write $x = K_0 z + h y$. In retained/pivot order, this transformation has determinant $(\det C_p)^{-1}$, and $Cx = y$. For the second, the normalized full Gaussian of covariance $Q_b^{-1}$ sends $Cx$ to a Gaussian of covariance $M_b$; its density at zero gives the factor. Equivalently,

$$\det Q_b \, \det(C Q_b^{-1} C^T) = (\det C_p)^2 \det A_b. \tag{88}$$

The delta in (87) uses the target's relative Lie coordinate. The actual target measure is group Haar, whose density at that coordinate origin is the fixed $h_0$. It supplies no factor $j(\log W)$. If one instead uses induced Euclidean measure on $\ker C$, its coarea factor is precisely included in (87).

For a scalar expression $F$, write $\mathcal{N}_0 F = F(W) - F(1)$. Combining (79), (85) and (87) proves the exact identity

$$\boxed{D_{M,K} = \mathcal{N}_0 \left[ \log\det G_b - \frac{1}{2} \log\det Q_b - \frac{1}{2} \log\det M_b \right].} \tag{89}$$

The pivot has disappeared, but the coarse-constraint determinant remains. Every completion allowed in (86) gives the same combined expression. The individual terms need not be completion-independent or locally gauge invariant.

At fixed regulator these are identities on a positive real germ and then on their chosen analytic branches. The algebraic identity (88) also continues wherever its inverses exist. No uniform complex domain for the three individual logarithms, as $K$ or volume grows, follows from the combined identity.

### 4.5 A positive completion from the actual minimum action

In the same relative target coordinate set

$$s_{\text{eff}}(y) = S_{MK}(e^{iy} W), \qquad q_{\text{eff}} = s''_{\text{eff}}(0).$$

The envelope identity gives $s'_{\text{eff}}(0) = \lambda$. The residual scalar

$$\mathcal{R}(x) = s(x) - s_{\text{eff}}(f(x))$$

is critical and nonnegative near zero by the actual local minimum comparison. It is invariant under the full local fine gauge group, because both scalar actions are locally gauge invariant and the composite map is equivariant. Its Hessian is therefore

$$H_{\text{res}} = \mathcal{L} - C^T q_{\text{eff}} C \geq 0, \qquad H_{\text{res}} D_f = 0 \tag{90}$$

for every fine infinitesimal gauge direction, with any convenient fixed global base convention. The nonnegativity uses the proved minimum comparison, not stationarity alone.

Let $H$ be the actual derivative of the minimum section in these relative coordinates. Since $CH = I$ and the residual vanishes identically along that section, its positive semidefinite Hessian obeys

$$H_{\mathrm{res}} H = 0, \qquad \mathcal{L} H = C^T q_{\mathrm{eff}}. \tag{91}$$

No independently chosen flat lift is substituted for $H$.

A canonical completion is

$$Q_b = H_{\mathrm{res}} + D_b D_b^T + C^T C > 0. \tag{92}$$

Indeed a null vector would satisfy $Cx = 0$, $D_b^T x = 0$ and $x^T \mathcal{L} x = 0$. The constrained quotient positivity then gives $x = D_b \alpha$, and $G_b \alpha = 0$ implies $x = 0$. This is (86) with $B = I - q_{\mathrm{eff}}$, which need not be positive itself. No uniform lower spectral bound for (92) is asserted as $K$ grows.

The general coarse compression can also be identified exactly. Set

$$H_b = H - D_b G_b^{-1} D_b^T H. \tag{93}$$

Then $CH_b = I$, $D_b^T H_b = 0$, $H_{\mathrm{res}} H_b = 0$, and $\mathcal{L} H_b = C^T q_{\mathrm{eff}}$. For every completion (86), it follows that

$$Q_b H_b = C^T (q_{\mathrm{eff}} + B), \qquad q_{\mathrm{eff}} + B = H_b^T Q_b H_b > 0. \tag{94}$$

Multiplication by the inverses gives

$$Q_b^{-1} C^T = H_b (q_{\mathrm{eff}} + B)^{-1}, \qquad \boxed{M_b = (q_{\mathrm{eff}} + B)^{-1}}. \tag{95}$$

Thus (89) is equivalently

$$D_{M,K} = \mathcal{N}_0 \left[ \log\det G_b - \frac{1}{2} \log\det Q_b + \frac{1}{2} \log\det(q_{\mathrm{eff}} + B) \right]. \tag{96}$$

The coarse covariance is the inverse of the actual relative-coordinate effective Hessian with its completion. For the canonical choice $B = I - q_{\mathrm{eff}}$, it is exactly $M_b = I$ and that coarse logarithm vanishes. The coarse physics remains inside $H_{\mathrm{res}} = \mathcal{L} - C^T q_{\mathrm{eff}} C$ and hence in $Q_b$; no uniform reference comparison is thereby obtained.

### 4.6 A soft based-ghost mode

At the flat background $D_b$ is the nearest-neighbor gauge gradient on scalar functions vanishing at the $K$-centers. Take $K \geq 3$ on a compatible unit-current torus and choose one real color. Let $\varphi$ be zero at all centers and one elsewhere. If there are $n_c$ centers, there are $n_f = K^d n_c$ current sites, and exactly $2d$ incident unit-difference bonds per center. Hence

$$\|\varphi\|_2^2 = (K^d - 1) n_c, \qquad \|D_b \varphi\|_2^2 = 2d n_c, \qquad \lambda_{\min}(G_b) \leq \frac{2d}{K^d - 1}. \tag{97}$$

This is an upper Rayleigh bound, not a matching spectral asymptotic. In four dimensions it implies

$$\|G_b^{-1}\| \geq \frac{K^4 - 1}{8}.$$

A final-physical-scale mass is $K^{-2}$ in the current coordinates. Thus the estimate $\|G_b^{-1}\| \leq C K^2$ is false uniformly in $K$. After resetting final spacing to one, $K^2 G_b$ still has a Rayleigh eigenvalue tending to zero.

This counterexample does not refute a bounded Maxwell finite-part difference for the complete expression (89). It rules out estimating the based ghost separately as a uniformly massive operator. The vector and coarse determinant must be kept coupled.

### 4.7 An explicit full-site and coarse Schur decomposition

Fix one final center as a global base. Let $D_f$ now denote the fine gauge generator for all scalar sites except that base, and split its domain into the non-$K$-centers and the other $K$-centers:

$$D_f = [\, D_b, D_e \,].$$

These are disjoint site-coordinate columns, not orthogonal images. The globally based action is free. Therefore

$$\begin{aligned} G_f &= D_f^T D_f > 0, \\ S_g &= D_e^T (I - D_b G_b^{-1} D_b^T) D_e > 0, \\ \det G_f &= \det G_b \, \det S_g. \end{aligned} \tag{98}$$

For any positive $Q_b$ as above, set

$$Q_f = Q_b + D_e D_e^T, \qquad S_v = I + D_e^T Q_b^{-1} D_e. \tag{99}$$

The determinant lemma gives

$$\det Q_f = \det Q_b \, \det S_v. \tag{100}$$

Consequently the actual correction is exactly

$$D_{M,K} = \mathcal{N}_0 \left[ \log\det G_f - \frac{1}{2} \log\det Q_f - \log\det S_g + \frac{1}{2} \log\det S_v - \frac{1}{2} \log\det M_b \right]. \tag{101}$$

$S_g, S_v$ are coarse-site matrices; $M_b$ is a coarse-bond matrix. The first two terms resemble a vector/ghost combination, but all coarse corrections remain. $G_f$ is globally based; it is not yet the translation-invariant all-site massive ghost of a reference calculation. If no nonbase coarse site remains, the corresponding zero-dimensional determinants are one. For the canonical completion $M_b = I$ by (95); the general formula retains all three coarse terms.

For (92), the vector operator is

$$Q_f = s'' + D_f D_f^T - \lambda \cdot f'' + C^T (I - q_{\text{eff}}) C. \tag{102}$$

If a proposed matched massive reference is $P_\eta = s'' + D_f D_f^T + \eta^2 I$, with $\eta = K^{-1}$, the exact difference is

$$Q_f - P_\eta = -\lambda \cdot f'' + C^T (I - q_{\text{eff}}) C - \eta^2 I. \tag{103}$$

The multiplier term vanishes at the flat background, but its first and second background derivatives can contribute to the determinant Hessian. Neither that vanishing nor the coarse rank bounds its Maxwell coefficient by itself. Equation (103) is a comparison interface, not an identification with an operator named in a different background-field convention. Any logarithmic comparison using $P_\eta$ additionally requires its invertibility on the chosen finite germ. No uniform such neighborhood is inferred here.

### 4.8 A uniformly bounded coarse effective-Hessian factor

The general identification in (95) permits a uniform estimate for one actual comparison term. Use the established strict scalar family of the entire action $S_{MK}$, including its pair-supported quadratic baseline. On a fixed small complex ball it has uniform spatial bounds and an arbitrary prescribed fixed cardinality reserve, independent of $MK$ and compatible volume.

For $W = e^{iw}$, set

$$\Xi_w(y) = \frac{1}{i} \log(e^{iy} e^{iw}).$$

This is a per-bond analytic map with uniformly bounded derivatives on a fixed small joint ball. With $E_w = D_y \Xi_w(0)$ and $U_w = D_y^2 \Xi_w(0)$,

$$q_{\text{eff}}(w)[u,v] = D^2 S_{MK}(w)[E_w u, E_w v] + DS_{MK}(w)[U_w(u,v)]. \tag{104}$$

The background-current term is retained. Compose each scalar support coefficient with $\Xi_w(y)$, differentiate twice in $y$, and up to twice in $w$ using fixed spare Cauchy radii. No support enlargement occurs. Rows, columns and background source slots stay in the same carrier. A fixed cardinality reserve absorbs their finite counting factors.

Consequently, for some fixed $\mu, \rho > 0$,

$$\sup_{\|w\|_\infty < \rho} \|q_{\text{eff}}(w)\|_\mu \le R_\mu, \tag{105}$$

where $\|\cdot\|_\mu$ is the maximum of the exponentially weighted absolute block row and column sums. The bound is uniform in $MK$ and volume. After reducing the exponent to $m > 0$, its source derivatives satisfy

$$\sum_{ij} e^{m[d(i,x)+d(j,x)]} |(\partial_x q_{\text{eff}})_{ij}| \le C_1,$$
$$\sum_{ij} |(\partial_x \partial_y q_{\text{eff}})_{ij}| \le C_2 e^{-md(x,y)}. \tag{106}$$

To obtain these bounds, split the carrier-tree exponential between its source separation and its row-to-source distance, then sum that distance. The coarse cell fiber has fixed dimension, with no factor from $K^4$ fine coordinates.

**Proposition 4.2** (Bounded actual coarse determinant contribution)**.** *Choose a fixed $a > 2R_\mu$ and the completion $B = aI$. The second-background-derivative kernel of*

$$\tfrac{1}{2}\mathcal{N}_0 \log\det(q_{\text{eff}} + aI)$$

*has a uniform exponential row bound, independent of the preceding depth, endpoint and compatible volume. Its true infinite Maxwell coefficient, and its actual finite reference-normalized coefficients above the reference threshold, have a uniform absolute bound.*

*Proof.* In the weighted Schur algebra, the Neumann series gives

$$P_c = q_{\text{eff}} + aI, \qquad \|P_c^{-1}\|_\mu \le 2/a. \tag{107}$$

The corresponding fine completion is positive on the real germ: write $x = \xi + H_b y$, $C\xi = 0$. Equation (94) makes the cross term zero, leaving the positive constrained-divergence form in $\xi$ and $y^T(q_{\text{eff}} + aI)y$. No uniform fine spectral gap is inferred.

At finite volume the series

$$\log\det P_c = (\dim P_c)\log a + \sum_{n\ge 1} \frac{(-1)^{n+1}}{n} \operatorname{Tr}(a^{-1} q_{\text{eff}})^n \tag{108}$$

converges absolutely. Let $K_c$ be the Hessian kernel of $\mathcal{N}_0 \log\det P_c$; the desired half-logarithm has Hessian $K_c/2$. Explicitly,

$$K_c(x,y) = \operatorname{Tr}\left[P_c^{-1} q_{\text{eff},xy} - P_c^{-1} q_{\text{eff},x} P_c^{-1} q_{\text{eff},y}\right]. \tag{109}$$

The first trace is bounded by the second line of (106) and the inverse bound. For the second trace, use the two source-rooted first derivative bounds and spend a spare inverse-edge exponent to connect their roots. Absolute summation gives

$$|K_c(x,y)| \le C_a e^{-m' d(x,y)} \tag{110}$$

for a fixed $m' > 0$, uniformly in depth and volume. Periodic sums are bounded by their infinite-lattice counterparts. Its second displacement moment is therefore uniformly bounded. The finite reference denominators are uniformly bounded below above their established threshold, which proves the finite statement as well. □

The actual action's local gauge identity makes its relative Hessian transform by orthogonal bond-frame conjugation under real local gauges. Thus the coarse massive determinant has the appropriate total local gauge identity on its common germ; no coefficientwise Ward identity was used in the proof. With $B = aI$, (96) now reduces the unbounded-endpoint comparison to the fine combination

$$\log\det G_b - \tfrac{1}{2}\log\det\left[s'' - \lambda\cdot f'' + D_b D_b^T + aC^T C\right],$$

up to the proved uniformly bounded coarse contribution. For a general completion the full-site vector difference is $-\lambda\cdot f'' + C^T B C - \eta^2 I$. For this fixed-$a$ choice it is

$$Q_f - P_\eta = -\lambda\cdot f'' + aC^T C - \eta^2 I. \tag{111}$$

Equation (103) instead records the canonical completion $B = I - q_{\mathrm{eff}}$. Each comparison uses all factors from its one chosen completion.

### 4.9 The remaining uniform comparison

The deposited outline asserts a bounded error in the Maxwell coefficient when its composite determinant is compared with a massive vector/ghost reference, at $\eta = \ell^{-k}$. The exact identities above do not establish that error bound. In particular, the formal background effective-action convention and massless nonzero-external-momentum amplitudes of a reference calculation do not identify this constrained normalizer automatically.

For a precisely matched reference, after removing the bounded coarse effective-Hessian contribution, the full-site formulation must still control the Maxwell moment of the coarse-site package

$$-\log\det S_g + \tfrac{1}{2}\log\det S_v,$$

together with the global-base and ghost-mass correction, and the Maxwell moment of $\log\det Q_f - \log\det P_\eta$ with all multiplier, coarse-completion and mass terms for the selected completion. They must also identify the reference background with the actual smooth pre-gauge minimum and preserve the actual finite extraction and zero-mode conventions. Based-gauge covariance allows evaluating (89) at that pre-gauge representative, but does not replace it by an arbitrary external interpolation.

Each determinant Hessian has the concrete formula

$$D^2\log\det P[u,v] = \mathrm{Tr}\left(P^{-1}P_{uv} - P^{-1}P_u P^{-1}P_v\right). \tag{112}$$

The established fixed-$K$ bounds make the relevant second-displacement-moment trace sums meaningful. Their constants may depend on $K$; they do not prove an $O(1)$ bound as $K\to\infty$. The soft mode (97) shows why a separate uniform inverse estimate cannot supply this step. A coupled cancellation or a scale-resolved comparison is still required. No universal coefficient, sign, averaged drift or ultraviolet coupling trajectory is deduced here.

# 5 The actual leading determinant correction satisfies a cocycle

This section proves an exact identity of normalized leading stationary amplitudes at a fixed finite regulator. It compares these amplitudes through one auxiliary local orbit-tube integral. It does not identify direct and sequential sharp-window sectors at positive coupling. All maps, pivot factors and retained Hessians are those of the actual composite ordinary constraint.

## 5.1 Objects and statement

Fix a finite compatible periodic regulator, a preceding homogeneous depth $M$, and two allowed ratios $R = \ell^r$, $L = \ell^s$. Let $X, Y, Z$ be the current, $R$-coarse and $RL$-coarse link configuration spaces, each with product normalized Haar measure. The actual iterated ordinary maps satisfy

$$f = F_R : X \longrightarrow Y, \qquad h = F_L : Y \longrightarrow Z, \qquad h \circ f = F_{RL}. \tag{113}$$

No equality with a direct-$R$ or direct-$RL$ ordinary average is used.

The coupling-free action $S_M$ is the established homogeneous ordinary minimum action, normalized at the identity. For an allowed ratio $K$, write $V_*^{M,K}(W)$ for the physical gauge-zero stationary representative, using the independently chosen direct-$K$ internal contour gauge $P_K$. Let $A_{M,K}(W)$ be the full retained Hessian of $S_M + G_K$ there, in the actual constraint chart. Let $d_K(V_*)$ be the product of inverse pivot Jacobians. Define

$$D_{M,K}(W) = \log \frac{d_K(V_*^{M,K}(W))}{d_K(1)} - \frac{1}{2} \log\det\Big[A_{M,K}(W) A_{M,K}(0)^{-1}\Big]. \tag{114}$$

Here $D_{M,K}(1) = 0$. These are the actual $\Delta + \mathcal{Q}_{\det}$ terms, with positive real branches near the identity and their given analytic continuations. Constraint curvature and the absolute contour-gauge Hessian are included in $A_{M,K}$.

Let $V_p^{MR,L}(W)$ be the outer pre-root-gauge stationary representative, and let $\mathcal{T}_{MR,L}$ denote substitution into this field.

**Theorem 5.1** (Finite leading-amplitude cocycle)**.** *On a real small-field germ,*

$$D_{M,RL} = \mathcal{T}_{MR,L} D_{M,R} + D_{MR,L}. \tag{115}$$

*The identity extends to every connected common analytic domain already available for its two sides. The germ used in its proof may depend on the fixed regulator and the ratios.*

The proof keeps all Haar, gauge and Gaussian normalization factors. They disappear only when each leading amplitude is divided by its value at the identity.

## 5.2 Based gauge groups and an exact insertion

Let $H_R$ consist of current site gauges equal to the identity at every intermediate $R$-center, and let $H_{RL}$ be based at the final $RL$-centers. Restriction to intermediate centers gives

$$1 \longrightarrow H_R \longrightarrow H_{RL} \longrightarrow H_L \longrightarrow 1, \tag{116}$$

where $H_L$ is the intermediate group based at final centers. The site coordinates split into the non-intermediate sites and the intermediate non-final centers. Embed $H_L$ by setting its other current-site values equal to the identity. The two factors have disjoint supports, so their direct product is $H_{RL}$ and normalized Haar is their product Haar. There is no background-dependent orbit-volume factor.

Each based action is free. A gauge preserving every bond of a connected graph is determined by its value at one site; its value at a based center is the identity. The actual maps obey

$$f(U^u) = f(U)^{u|_{\text{intermediate centers}}}, \qquad h(V^v) = h(V)^{v|_{\text{final centers}}}. \tag{117}$$

Thus $f$ is $H_R$-invariant, $h$ is $H_L$-invariant, and $h \circ f$ is $H_{RL}$-invariant. Product link Haar is invariant under all these site actions.

For $u \in H_K$, endpoint covariance of the actual direct-$K$ contour variables gives

$$P_K(U^u; y, x) = P_K(U; y, x)u(x)^{-1}.$$

With the actual nonnegative class penalty $q(v) = 1 - \operatorname{Re}\tau(v)$, whose unique zero is the identity, put $G_K(U) = \sum_{x \text{ not a center}} q(P_K(U; y, x))$. Then

$$\int_{H_K} e^{-G_K(U^u)/g^2}\, du = c(g)^{n_{s,K}}, \qquad c(g) = \int_G e^{-q(v)/g^2}\, dv. \tag{118}$$

It is therefore exact to insert $e^{-G_K/g^2}$ into an $H_K$-invariant tube integral and multiply by $c(g)^{-n_{s,K}}$. This factor is independent of the target field. Its asymptotic constant and power of $g$ need not be computed below.

### 5.3 A consistent local orbit-tube phase

The original $S_M$ is known on a small-link domain. We first justify extending it to the particular saturated tube needed for an exact pushforward composition.

Suppose two sufficiently small configurations $U, V$ satisfy $V = U^u$ with $u \in H_{RL}$. The bond equation, in a unitary-invariant matrix norm, yields

$$\|u(x) - u(x + e_\mu)\| \le C(\|U(x, \mu) - I\| + \|V(x, \mu) - I\|). \tag{119}$$

Every site has a path of at most $C_d RL$ edges to a final center, where $u = I$. Consequently

$$\|u - I\|_\infty \le C_d RL(\|U - I\|_\infty + \|V - I\|_\infty). \tag{120}$$

Shrink the comparison ball so that the near-identity logarithm $\lambda = \log u$ exists. Its edge differences satisfy the same type of estimate as (119). Along $u_t = \exp(t\lambda)$ the transformed links stay within a fixed multiple of the endpoint smallness. Fixed color-norm conversions are included in this constant. Shrinking the ball again keeps the whole path in the established action domain. Local Ward invariance therefore gives

$$S_M(U) = S_M(V). \tag{121}$$

The same argument applies on the intermediate graph and to $H_L$. No radius uniform in $RL$ is claimed in this step.

On the $H_{RL}$-saturation of this small representative neighborhood, define

$$S_{\text{ext}}(U^u) = S_M(U). \tag{122}$$

Equation (121) makes the definition independent of the small representative. The free compact group action has an actual local slice $P_{RL} = I$: the unique based transformation to it is $u(x) = P_{RL}(U; y, x)$. Hence $S_{\text{ext}}$ is smooth, and real analytic in local slice coordinates, and is invariant on the saturation.

The local ordinary maps and contour variables extend equivariantly to this tube. One may use their endpoint-covariant relative-log formulas or define them from the slice. On overlaps the two definitions

agree by (117) and the original local formulas. No measurable completion away from the tube is invoked.

The proved physical branch is a strict local constrained minimum modulo the based orbit. Its actual retained Hessian after adding $G_K$ is positive. On the slice $P_K = I$, the gauge penalty is identically zero, so that positivity also gives positivity of the action Hessian on the quotient constrained tangent. The finite-dimensional implicit-function argument yields a unique local critical orbit in a smaller neighborhood.

Choose a compact closure of a sufficiently small real target germ. Its stationary orbits form a compact set. Choose a smooth $H_{RL}$-invariant cutoff $\chi$, equal to one near these orbits and supported in a tube containing no other quotient saddle. It can be chosen one near every inner conditional minimum over a smaller compact outer saddle neighborhood as well: use continuity of the minimum maps, choose nested neighborhoods, and average the cutoff over the compact based action. The averaging leaves it one on the invariant inner neighborhood. The quotient phase has a positive gap on the remaining compact part of the support.

This cutoff is an auxiliary device for determining a leading coefficient. It is neither the actual retained sharp window nor a proposed global Wilson density.

### 5.4 Exact Haar pushforward composition

For $g > 0$, consider the positive finite measure

$$\chi(U)e^{-S_{\mathrm{ext}}(U)/g^2}\, d\mu_X(U). \tag{123}$$

The maps $f, h, h \circ f$ are submersions near all relevant orbits. At a small representative, the actual pivot blocks give full rank; equivariance preserves rank on the orbit. A smaller tube retains the property. Its local pushforward densities are thus smooth relative to product Haar.

Write $J_g(V)$ for the pushforward density through $f$, and $I_g(W)$ for the density through $h \circ f$. Exact pushforward composition gives

$$I_g(W) = \int_{h(V)=W} J_g(V)\, d\mu_h(V). \tag{124}$$

Here $d\mu_h$ is the conditional coarea density relative to product Haar. Equivalently, test both sides against a smooth function of $W$ and use ordinary Fubini. No multiplication of formal delta distributions is required.

The first pushforward is exactly $H_L$-invariant. Indeed, lift a based intermediate gauge by (116), use the $H_{RL}$ invariance of $\chi$ and $S_{\mathrm{ext}}$, and change the current Haar variables. This is precisely the invariance needed in the outer integral.

The actual two-minimum comparisons identify the inner critical value and then the outer one:

$$\min_{f(U)=V} S_{\mathrm{ext}}(U) = S_{MR}(V), \qquad \min_{h(V)=W} S_{MR}(V) = S_{MRL}(W). \tag{125}$$

These are local comparisons on the chosen tube. The conditional minima depend smoothly on the outer parameters modulo the inner based orbit. The cutoff is one there. Outside the smaller outer saddle neighborhood, the compact quotient phase gap makes the contribution exponentially smaller. All uniformity here concerns the fixed finite regulator and fixed ratios only.

### 5.5 The finite-dimensional Laplace fact

We record the elementary asymptotic used twice. In a finite real chart suppose

$$F(z, w) = s(w) + \tfrac{1}{2} z^T A(w) z + O(|z|^3), \qquad A(w) > 0,$$

and the critical point is the unique local minimum on the chosen compact support. The density $m(z,w)$ is smooth and its cutoff is one near zero. On a small compact parameter germ the Hessian has a common positive lower bound. Then

$$\int e^{-F(z,w)/g^2} m(z,w)\,dz = e^{-s(w)/g^2} g^n (2\pi)^{n/2} m(0,w) \det A(w)^{-1/2}[1+o(1)] \tag{126}$$

uniformly in that parameter germ.

To prove it, shrink the chart until $F - s \geq c|z|^2$. Set $z = g\xi$, first restrict $\xi$ to a fixed bounded set, and use Taylor's formula for uniform convergence to the Gaussian. A common Gaussian majorant removes that restriction. The compact remainder outside a smaller chart has a positive phase gap and is exponentially negligible. This proof also allows a smooth positive amplitude, whose leading value is its value at the saddle, and an amplitude family converging uniformly there with a bounded relative error.

For an orbit integral, first insert (118). The phase becomes $S + G_K$. Its unique saddle is the gauge-zero representative: the quotient phase has its strict minimum there modulo the orbit, and on the compact based orbit $G_K$ has its unique zero at $P_K = I$. Away from a neighborhood of that representative, the sum of the quotient and gauge penalties has a positive gap. The actual retained chart therefore reduces the leading asymptotic to (126), with the exact factor $c(g)^{-n_{s,K}}$ still present. No unspecified quotient volume form or ultraviolet-uniform error estimate is used.

### 5.6 Identification with the actual pivot and determinant

The exact local constrained measure has, at its saddle, the factor

$$h_0^{n_e - n_c} d_K(V_*). \tag{127}$$

Indeed relative Haar coordinates are $d(e^{iB}) = h_0 j(B)\,dB$ with $j(0) = 1$; the normalized fluctuation straightening Jacobian is one at zero; and solving the constraint contributes the inverse pivot product. The target density is relative to group Haar. Consequently there is no extra factor $j(\log W)$ in (127).

Let $I_{M,K;g}$ denote an appropriate local tube pushforward. The phase-corrected, identity-normalized stationary amplitude is

$$e^{D_{M,K}(W)} = \lim_{g \downarrow 0} e^{[S_{MK}(W) - S_{MK}(1)]/g^2} \frac{I_{M,K;g}(W)}{I_{M,K;g}(1)}. \tag{128}$$

The gauge integral, powers of $g$, Haar constants, flat determinants and all scalar cutoff factors cancel in this ratio. The remaining saddle measure and Hessian are exactly (114). A cutoff equal to one near the critical orbit changes no leading coefficient.

The actual sharp-window integral has the same saddle, pivot factor and retained Hessian. Its variable is $z = g\xi$ and its window is the actual product window of threshold $T(g)$. Choose $T(g) \to \infty$ with $gT(g)^3 \to 0$. The established centered Gaussian estimate makes its normalized logarithmic expectation

$$O\Big(\tau(T(g)) + g^2 T(g)^6\Big) \longrightarrow 0. \tag{129}$$

Thus the sharp-window and the auxiliary fixed-tube integrals have the same normalized leading amplitude. This does not equate the integrals at positive $g$.

The auxiliary first pushforward supplies only the intermediate based $H_L$ invariance: its input was made $H_{RL}$-invariant. The phase and normalized leading amplitude therefore have that identity on the needed intermediate orbit. It suffices to replace the rooted outer saddle by $V_p$. The cutoff has not been made invariant under every output site gauge with nontrivial final-center values. The full local output gauge identity for the actual $D$ is the previously proved one from the zero-input quantum leading limit, not an additional invariance claimed for this cutoff.

### 5.7 Sequential saddle evaluation

*Proof of Theorem 5.1.* Apply (126) on the $f$ fibers, uniformly over a compact neighborhood of the outer stationary orbit. With a scalar $c_1(g)$ independent of $V$,

$$J_g(V) = c_1(g)e^{-S_{MR}(V)/g^2}e^{D_{M,R}(V)}[1 + o(1)]. \tag{130}$$

The relative error is uniform. Exact $H_L$ invariance reduces the full compact outer orbit to a small slice. The inner cutoff is one at every relevant conditional minimum.

Insert (130) into (124). The outer Laplace formula evaluates the smooth amplitude $e^{D_{M,R}(V)}$ at its existing classical minimum. With another scalar $c_2(g)$,

$$I_g(W) = c_1(g)c_2(g)e^{-S_{MRL}(W)/g^2}e^{D_{M,R}(V_p^{MR,L}(W))+D_{MR,L}(W)}[1 + o(1)]. \tag{131}$$

The amplitude gradient has no factor $g^{-2}$ and hence does not alter the leading phase saddle. The outer rooted representative gives the same amplitude by the based $H_L$ invariance. The compact phase gap controls the remaining part of the outer integration.

Direct evaluation of the same integral through $h \circ f$ instead gives a scalar $c_{12}(g)$ and

$$I_g(W) = c_{12}(g)e^{-S_{MRL}(W)/g^2}e^{D_{M,RL}(W)}[1 + o(1)]. \tag{132}$$

Normalize both equations by their values at $W = I$ after removing the phase. There $V_p = I$ and every $D$ is zero. All scalar factors cancel, without any computation of the relation between $c_{12}$ and $c_1c_2$. Taking $g$ to zero yields

$$e^{D_{M,RL}(W)} = e^{D_{M,R}(V_p^{MR,L}(W))+D_{MR,L}(W)}.$$

On the positive real germ the logarithms are real and zero at the identity. They therefore satisfy (115), without a branch ambiguity. Both sides are given holomorphic pivot/determinant functionals. The identity theorem extends their germ identity to each connected common analytic domain. The smaller comparison germ used by Laplace does not shrink a domain already established for those functions. □

### 5.8 Joint infinite systems and the Maxwell coefficient

For fixed $R, L$, the existing modules give uniform anchored analytic families for all three $D$ terms. On a joint local limit system the finite identity passes on compact fields by local normal convergence and anchored tail bounds; the same bounds justify the classical substitutions. It gives a scalar quotient cocycle, not a coefficientwise identity of carrier decompositions.

Apply exact preservation of the true infinite Maxwell coefficient under classical transport. Then

$$\beta_{MRL}(D_{M,RL}) = \beta_{MR}(D_{M,R}) + \beta_{MRL}(D_{MR,L}). \tag{133}$$

Thus a long fixed-ratio leading coefficient is the sum of its consecutive step coefficients in this joint system. This proves neither their sign nor their convergence as the preceding depth grows.

At finite volume, the symmetric short-cutoff moment $\beta_{M,N}$ does not commute exactly with classical transport. One must retain the separately controlled finite leakage when applying it to (115). No exact finite scalar $\beta$ additivity is asserted.

### 5.9 Scope

The result rests on a consistent local orbit-tube extension, exact finite Haar pushforward composition, and ordinary finite-dimensional saddle asymptotics using the actual pivot, contour penalty and retained Hessian. It assumes no ultraviolet-uniform stationary-phase remainder.

It does not identify direct and sequential sharp-window sectors at positive $g$, extend the original local action to every gauge configuration, or identify every local sector with the full Wilson measure. The auxiliary cutoff disappears only from the leading coefficient.

The background-field/ghost determinant comparison, a bounded comparison error in depth, the numerical logarithmic coefficient, its sign, physical coupling tuning and the treatment of the other field sectors remain separate from this cocycle.

# 6 Qualitative transfer of a positive leading block drift

The compactness and moment-passage inputs in this section are available for the actual homogeneous local macrostep. Its positivity hypothesis is explicitly additional and unproved. The conclusion requires no quantitative finite-volume rate.

## 6.1 The actual fixed-length coefficients

Fix the primitive ratio, elementary composite ratio $R$, grouping $J$, common analytic domains and extraction convention. Fix a finite $q \geq 1$, chosen to be a multiple of the activity-envelope block length $q_a$ when applying the tuning consequence below. Let $\mathcal{P}$ be the compatible nested cubic period family: every fixed power of $R$ divides all sufficiently late periods in $\mathcal{P}$.

For a $q$-step block with final period $N \in \mathcal{P}$, put

$$M_j = MR^j, \qquad Q_j = R^{q-j}N, \qquad 0 \leq j \leq q. \tag{134}$$

Here $M$ is any permitted preceding homogeneous depth. Let $D_{M_j,R;Q_{j+1}}$ be the actual finite leading correction $\Delta + \mathcal{Q}_{\mathrm{det}}$ of step $j$. Define

$$b_{0,j}^{M,N} = \beta_{M_jR,Q_{j+1}}(D_{M_j,R;Q_{j+1}}), \qquad B_q(M,N) = \sum_{j=0}^{q-1} b_{0,j}^{M,N}. \tag{135}$$

Each functional is the actual finite short-lift ratio normalized by its actual finite classical reference. These coefficients are real on the real branches. No finite ratio is replaced by an infinite moment.

The uniform outer-domain leading-family bound and the finite moment estimate give, above the fixed reference threshold,

$$|b_{0,j}^{M,N}| \leq B_0 \tag{136}$$

uniformly in depth and permitted volume. The reference denominators are bounded below by $1/2$.

## 6.2 Varying-depth structural limits and their moments

Take any $M_n$ and any $N_n \in \mathcal{P}$ tending to infinity. The old depths need not be bounded or converge. Extract jointly the classical references and background maps at the shifted depths $M_nR^j$, their required fixed-order vertices, the actual retained flat precisions and covariances, and all $q$ leading corrections in (135). The uniform strict support, common-domain and exponential source-Hessian bounds provide such a subsequence. If a structural reference comparison requires additional fixed depth shifts or endpoint powers, include their countable collection in the same diagonal extraction. Cofinal divisibility guarantees their eventual finite availability.

Write the limiting corrections as $D_j^*$, with true infinite Maxwell functional $\beta_{j+1}^*$ at output scale $j+1$, and set

$$B_q^* = \sum_{j=0}^{q-1} \beta_{j+1}^*(D_j^*). \tag{137}$$

The scale indices describe the structural limit. They do not identify it with one fixed finite-depth system.

These are actual determinant structural limits. At fixed $R$ the retained fiber per current block has fixed finite dimension; the whole retained operator still has dimension growing with volume. Local matrix convergence and exponential envelopes pass products and the inverse identity $A_0C_0 = I$ by truncating the intermediate sums. The chart and pivot maps are fixed; common-domain Cauchy estimates give convergence of classical vertices through order four. The absolutely convergent trace formula therefore identifies the limiting Hessian with the same actual determinant expression. One may equivalently extract the strict scalar families jointly and identify their Hessians in this way. No old fine coordinate count occurs.

**Lemma 6.1** (Passage of the actual finite moments)**.** *Along every such joint subsequence,*

$$b_{0,j}^{M_n,N_n} \longrightarrow \beta_{j+1}^*(D_j^*) \quad (0 \le j < q), \qquad B_q(M_n, N_n) \longrightarrow B_q^*. \tag{138}$$

*Proof.* For a fixed $j$, the leading source-Hessian kernels converge locally and have a common exponential row bound. Restrict their short-lift second moment first to a fixed displacement radius. The finite sum converges. The rest has a uniformly summable exponential second-moment tail. The finite short-lift cutoff tends to infinity since $Q_{j+1} = R^{q-j-1}N_n \to \infty$.

The actual finite reference denominator tends uniformly to one by the flat normalized classical periodization estimate, even for these varying depths. Division proves the first assertion. The sum has fixed finite length $q$, proving the second. Neither an extensive total limit nor a regulator convergence rate was used. □

### 6.3 The conditional positive transfer

**Unproved structural positivity hypothesis.** Assume there is one $\delta > 0$ such that *every* joint structural infinite limit obtainable above satisfies

$$B_q^* \ge \delta. \tag{139}$$

The same margin must work for all varying-depth and permitted-period sequences. Positivity in one selected fixed-depth limit is insufficient.

**Proposition 6.2** (Uniform finite block lower bound)**.** *Under* (139)*, for every* $0 < \varepsilon < \delta$ *there is a threshold* $N_*(\varepsilon)$*, independent of preceding depth, such that*

$$B_q(M, N) \ge \delta - \varepsilon \quad \textit{for all permitted } M, \quad N \in \mathcal{P}, \quad N \ge N_*(\varepsilon). \tag{140}$$

*In particular the finite block is at least* $\delta/2$ *above one uniform period threshold.*

*Proof.* Otherwise choose $N_n \to \infty$ in $\mathcal{P}$ and old depths $M_n$ such that $B_q(M_n, N_n) < \delta - \varepsilon$. Take a joint structural subsequence. By Lemma 6.1, its limiting sum is at most $\delta - \varepsilon$, contradicting (139). □

The threshold is also uniform in a block's starting stage: absorb that stage into its preceding depth $M$. Its final period is the $N$ in (134). The proof gives no numerical threshold or convergence rate.

### 6.4 The finite tuning input needs only this qualitative bound

Take $q$ to be a multiple of $q_a$. Write $\mathcal{F}_{j,g}$ for the actual finite quantum extraction map, $x_j$ for the actual real extracted input class, and $d_j(g, x) = \beta_{j+1}(\mathcal{F}_{j,g}(x))$ for its inverse-coupling decrement. On the established finite activity envelope the actual decrement obeys

$$\begin{aligned} |d_j(g, x_j) - b_{0,j}| \le{}& C_{\mathrm{int}}\omega(N_{\min}) + C_{\mathrm{par},\beta}\{\tau(T(g)) + g^2T(g)^6\} \\ &+ C_{\mathrm{in},\beta}\, gT(g)/b. \end{aligned} \tag{141}$$

The first term tends qualitatively to zero as the period threshold increases. The other terms tend to zero in the weak-coupling limit at the fixed field radius and chosen real threshold $T(g)$.

Use (140) with $\varepsilon = \delta/2$. Raise the period threshold and lower the weak-coupling ceiling until (141) is at most $\delta/(4q)$. Then every permitted $q$-step block satisfies

$$\frac{\delta}{4} \le \sum_{j=k}^{k+q-1} d_j(g_j, x_j) \le qD_{\mathrm{abs}}. \tag{142}$$

The upper bound is the proved absolute-increment bound. Repeating the $q_a$-step activity blocks gives the same intermediate envelope throughout this longer block. No new large-endpoint field-dilation estimate is used.

These are the positive block bounds used in the continuous clipped-recursion shooting argument. Put $\Gamma_{\mathrm{w}} = g_{\mathrm{w}}^{-2}$, where $g_{\mathrm{w}}$ is the chosen weak-coupling ceiling. For a terminal inverse coupling $\Gamma_* \ge \Gamma_{\mathrm{w}} + qD_{\mathrm{abs}}$, that argument then gives an initial value in

$$[\,\Gamma_* + m\delta/4,\ \Gamma_* + mqD_{\mathrm{abs}}\,] \tag{143}$$

which makes $mq$ genuine weak local extraction steps and reaches $\Gamma_*$. This remains conditional on (139).

If an averaged formulation for arbitrary finite interval lengths is desired, write $n = mq + r$, $0 \le r < q$, and use (136) on the final fringe. For intervals entirely above the common period threshold,

$$\sum_{j=k}^{k+n-1} b_{0,j} \ge cn - C_{\mathrm{fin}}, \qquad c = \frac{\delta}{2q}, \qquad C_{\mathrm{fin}} = (q-1)(B_0 + c). \tag{144}$$

Indeed the complete blocks contribute at least $m\delta/2$, and the fringe contributes at least $-rB_0$. The shooting proof needs only the fixed-block bound, so this additional formulation is optional.

### 6.5 The remaining sign problem

Include the one long-ratio $R^q$ determinant in the joint extraction. The proved leading cocycle and true infinite classical marginal preservation give

$$B_q^* = \beta_q^*(D_{0,R^q}^*). \tag{145}$$

A uniformly positive reference comparison for this coefficient in every structural limit would therefore supply (139). The equality itself proves no sign.

Once that positive structural margin is established, qualitative compactness is enough for the needed finite block lower bound; a separate exponential regulator rate is unnecessary. This section does not establish the margin, exact finite marginal additivity, summability of errors over an infinite history, uniqueness of the old-depth limit, or convergence of tuned local densities to a continuum measure.

## 7 Finite ultraviolet tuning under an averaged drift hypothesis

The finite activity induction, leakage estimate, zero-input parity remainder and real-coupling continuity are available. This section gives a conditional endpoint argument for their actual finite local maps. Its additional hypothesis is a uniformly positive averaged drift of the *actual finite* leading determinant coefficients. That hypothesis is not proved here.

### 7.1 The actual decrement on the induction envelope

Fix the finite-induction data: elementary ratio $R$, grouping $J$, radius $b$, carrier exponent $a_0$, amplitude block length $q_a$, endpoint radius $r_b = K_{\mathrm{amp}} b^2$, and intermediate bound $C_{\mathrm{int}} b^2$. All references and $\beta_j$ are the actual finite reference-normalized ones. For a real extracted input define

$$\begin{aligned} d_j(g,x) &= \beta_{j+1}(F_{j,g}(x)), \\ \gamma_{j+1} &= \gamma_j - d_j(g_j,x_j), \qquad g_j = \gamma_j^{-1/2}, \\ x_{j+1} &= P_{j+1} F_{j,g_j}(x_j). \end{aligned} \tag{146}$$

Real branches and real initial data give real decrements. The activity induction applies to every realized weak coupling sequence while the permitted periods stay above its threshold. In particular $x_0 = 0$ is allowed.

Write

$$b_{0,j} = \beta_{j+1}(D_{M_j,R;N_{j+1}}) \tag{147}$$

for the finite pivot/Gaussian leading correction. It is independent of $g$ and of the inherited interaction, but depends on the actual classical data and regulator. It is not replaced by an independently selected infinite coefficient.

Since every $x_j$ is extracted, the finite leakage bound, parity estimate and quantum input comparison give

$$\begin{aligned} |d_j(g,x_j) - b_{0,j}| \le{}& C_{\mathrm{int}}\omega(N_{\min}) + C_{\mathrm{par},\beta}\{\tau(T(g)) + g^2 T(g)^6\} \\ &+ C_{\mathrm{in},\beta}\frac{gT(g)}{b}. \end{aligned} \tag{148}$$

The first term is the finite classical leakage, the second includes the fixed outer-radius factor, and the third includes $C_{\mathrm{int}}$ and the beta bound. This follows from $F = T + G$ and $G(x) = G(0) + (G(x) - G(0))$. The $b^2$ in the inner-radius remainder cancels beta's $b^{-2}$ bound. No exact finite identity $\beta_{\mathrm{out}} T = \beta_{\mathrm{in}}$ is used.

At fixed $b, R, J$, any prescribed tolerance $\epsilon > 0$ in (148) can be obtained by raising one period threshold and then lowering a coupling ceiling. The choices are independent of the number of permitted steps. The finite-induction magnitude bound also supplies

$$|d_j(g,x_j)| \le D_{\mathrm{abs}} \tag{149}$$

uniformly on the envelope and in the allowed history lengths and regulators.

### 7.2 The additional hypothesis and positive block increments

**Unproved averaged-drift hypothesis.** Assume that there are constants $c > 0$ and $C_{\mathrm{fin}} \ge 0$, independent of old depth and the permitted finite periods, such that

$$\sum_{j=k}^{k+n-1} b_{0,j} \ge cn - C_{\mathrm{fin}} \tag{150}$$

on the contemplated stage intervals. It suffices to assume this for the block length and starting stages used below. No pointwise positivity of an individual coefficient is required.

Choose $q$ to be a positive multiple of $q_a$ with $C_{\mathrm{fin}} \le cq/4$. Strengthen the period and coupling thresholds until the error in (148) is at most $c/4$. Every $q$-step sequence of projected maps started in the endpoint ball, at arbitrary prescribed couplings in that weak range, then satisfies

$$B_- \le \sum_{j=k}^{k+q-1} d_j(g_j,x_j) \le B_+, \qquad B_- = \frac{cq}{2} > 0, \qquad B_+ = qD_{\mathrm{abs}}. \tag{151}$$

Indeed the lower bound is $cq - C_{\mathrm{fin}} - q\epsilon \geq cq/2$, and the upper bound follows from (149).

Each consecutive $q_a$ subblock returns to the same endpoint ball. Thus the same $C_{\mathrm{int}}b^2$ envelope controls the entire larger block. This step repeats already proved quantum blocks; it does not invoke a new direct large-endpoint high-degree estimate or require a new field radius depending on $q$.

### 7.3 A continuous auxiliary clipped recursion

Let $\Gamma_{\mathrm{w}} = g_{\mathrm{w}}^{-2} > 0$ be the strengthened weak threshold. For the shooting proof only, evaluate the map for every real scalar $\gamma$ at

$$g_{\mathrm{eval}}(\gamma) = \max\{\gamma, \Gamma_{\mathrm{w}}\}^{-1/2}, \tag{152}$$

while leaving the scalar update unclipped:

$$\begin{aligned} \gamma_{\mathrm{next}} &= \gamma - d_j(g_{\mathrm{eval}}(\gamma), x), \\ x_{\mathrm{next}} &= P_{j+1}F_{j,g_{\mathrm{eval}}(\gamma)}(x). \end{aligned} \tag{153}$$

This is auxiliary when $\gamma < \Gamma_{\mathrm{w}}$; no physical density interpretation is assigned there. Its evaluated couplings always belong to the proved weak range. Starting in the endpoint ball, finite induction keeps all activity arguments admissible independently of the scalar value.

Joint real-coupling and quotient continuity, together with continuity of the maximum in (152), makes every finite clipped history continuous in its initial $\gamma$. On a compact initial interval, (149) bounds all intermediate scalars. Hence their evaluated couplings belong to a compact positive interval. Neither continuity at zero coupling nor a continuous representative selection nor a volume-uniform modulus is needed.

Since (148) and (150) apply to every admissible realized weak sequence, the clipped history also obeys (151).

### 7.4 Endpoint tuning and removal of clipping

**Theorem 7.1** (Conditional finite ultraviolet tuning)**.** *Assume* (150) *with the uniformity just specified. Fix $m \geq 1$ and a permitted finite history of $mq$ steps whose final period is above the strengthened threshold, with the nested-family compatibility conditions. Fix a terminal inverse coupling satisfying*

$$\Gamma_* \geq \Gamma_{\mathrm{w}} + qD_{\mathrm{abs}}. \tag{154}$$

*For $x_0 = 0$, or any fixed real extracted input in the endpoint ball, some initial inverse coupling in*

$$[\,\Gamma_* + mB_-,\ \Gamma_* + mB_+\,] \tag{155}$$

*generates $mq$ genuine weak local extraction steps and ends exactly at $\Gamma_*$. The actual scalar relation is $\gamma_{j+1} = \gamma_j - \beta_{j+1}(F_{j,g_j}(x_j))$, with $g_j = \gamma_j^{-1/2}$ throughout.*

*Proof.* Run the clipped recursion with the fixed initial activity. Summing (151), the final scalar at the left endpoint of (155) is at most $\Gamma_*$, while at the right endpoint it is at least $\Gamma_*$. Continuity and the intermediate value theorem give a clipped history ending at $\Gamma_*$.

For this selected history, sum the positive *future* block decrements to obtain

$$\Gamma_* + (m-k)B_- \leq \gamma_{kq} \leq \Gamma_* + (m-k)B_+, \qquad 0 \leq k \leq m. \tag{156}$$

Within any block, (149) implies

$$\gamma_{kq+r} \geq \gamma_{kq} - rD_{\mathrm{abs}} \geq \Gamma_* - qD_{\mathrm{abs}} \geq \Gamma_{\mathrm{w}}, \qquad 0 \leq r \leq q. \tag{157}$$

Thus clipping never occurred on the selected history. Its scalar and activity updates are the actual local extraction updates, and its coupling stays weak. The exact finite representative and vacuum compensation identities implement the same normalized local density at every step. □

The initial bare coupling therefore obeys

$$(\Gamma_* + mB_+)^{-1/2} \le g_{\text{initial}} \le (\Gamma_* + mB_-)^{-1/2}. \tag{158}$$

Conditional on the drift hypothesis, this tunes arbitrarily long finite histories to one fixed positive weak endpoint, with bare coupling of order $m^{-1/2}$. It proves no uniqueness of the tuned initial value or consistency between the choices at different $m$.

### 7.5 What remains unproved

The additional input is precisely (150) for the actual finite determinant coefficients with their actual extraction. The established activity bounds, parity, continuity and qualitative finite leakage do not prove it. A determinant cocycle can telescope the infinite counterpart, but a quantitative one-loop comparison and finite-regulator control must still yield the required positive averaged finite drift.

The horizons considered here are multiples of $q$, enough for an unbounded ultraviolet subsequence if that drift input is supplied. No assertion about every integer horizon is needed. The initial period grows to keep the final period above the common threshold. The theorem does not establish convergence of the tuned measures, mixed or large-field histories, a nontrivial continuum limit, or reconstruction. Its density statement remains the one for the specified homogeneous local extraction map and its justified continuation.

## 8 The remaining determinant estimate and the existence argument

The leading correction is now an actual functional,

$$D_{M,R} = \Delta_{M,R} + \mathcal{Q}_{\text{det},M,R},$$

with an exact Hessian trace formula and an exact finite composition law. In a jointly compatible infinite system its Maxwell coefficient satisfies

$$\beta_{MRL}(D_{M,RL}) = \beta_{MR}(D_{M,R}) + \beta_{MRL}(D_{MR,L}).$$

These identities supply the algebra for accumulated drift; they do not determine its value or sign.

The exact gauge and coarea bridge makes the comparison operators concrete. With its stated finite coordinates,

$$D_{M,K} = \mathcal{N}_0\left[\log\det G_b - \frac{1}{2}\log\det Q_b - \frac{1}{2}\log\det(CQ_b^{-1}C^T)\right],$$

where $\mathcal{N}_0$ subtracts the value at the identity. The inverse pivot has been accounted for by coarea. In fact $CQ_b^{-1}C^T = (q_{\text{eff}} + B)^{-1}$. Choosing one fixed sufficiently large $B = aI$, the coarse term $\frac{1}{2}\mathcal{N}_0 \log\det(q_{\text{eff}} + aI)$ has a uniformly bounded Maxwell coefficient by the proved coarse-action support bounds. The remaining comparison concerns the fine combination

$$\mathcal{N}_0\left[\log\det G_b - \frac{1}{2}\log\det(s'' - \lambda\cdot f'' + D_bD_b^T + aC^TC)\right].$$

In particular, the point-anchor Rayleigh test rules out a uniform physical mass estimate for the based ghost alone.

A concrete route is to compare the composite determinant from the bare Wilson action to a background-field vector and ghost reference at the same cutoff and infrared scale. With every normalization specified, the desired estimate has the form

$$|\beta(D_{1,R^k}) - B_k^{\text{ref}}| \le C, \qquad |B_k^{\text{ref}} - a\,k| \le C', \qquad a > 0,$$

with constants independent of $k$. Neither displayed bound is proved in this notebook. The stationary lift, constrained-current curvature and measure factors must be included in that comparison. A coefficient quoted for a different effective action does not establish these estimates for the present one.

The drift hypothesis required for shooting concerns the actual finite regulators, including arbitrary permitted preceding depths. An infinite comparison must therefore retain the uniformity and joint-limit compatibility needed to transfer a positive block margin to those finite coefficients. The qualitative transfer proposition supplies that passage once a positive margin holds in every relevant structural limit; a quantitative finite-volume convergence rate is unnecessary for this purpose. No exact finite commutation of Maxwell extraction with classical transport has been assumed.

Once that drift is supplied, the conditional shooting theorem closes the finite weak-coupling trajectory issue for this local map. The subsequent existence argument must still incorporate all required field histories with their exact measure identities, prove uniform source estimates, construct a nontrivial continuum observable hierarchy, and establish its Euclidean symmetry, positivity and reconstruction properties. These requirements are those of Notebook 01; the determinant and tuning arguments do not replace them.

## References


[1] *Toward a four-dimensional Yang–Mills construction*, research Notebooks 15–31 in this workspace. The local minimum, exact constrained integral, common-domain quantum map, finite induction and classical Maxwell transport are the principal inputs.

[2] M. Lüscher and P. Weisz, *Background field technique and renormalization in lattice gauge theory*, Nucl. Phys. B **452** (1995), 213–233. Primary preprint.

[3] M. Lüscher and P. Weisz, *Computation of the relation between the bare lattice coupling and the MS coupling in SU(N) gauge theories to two loops*, Nucl. Phys. B **452** (1995), 234–260. Primary preprint. These papers supply background-field formalism and reference amplitudes to investigate. They are not used here as a proof of the required comparison with our constrained determinant.

[4] E. Sussman and collaborators, Yang–Mills-LLM-experiment, deposited raw accounts at commit `e392ef45013a581664bab76b90867170b30fe10e`.